College of Engineering and Applied Sciences

Course Name: Lab. Intro. To Microprocessors

Course Number: ECE 2510

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Lab #4

Date: February 24rd 2026

Student:

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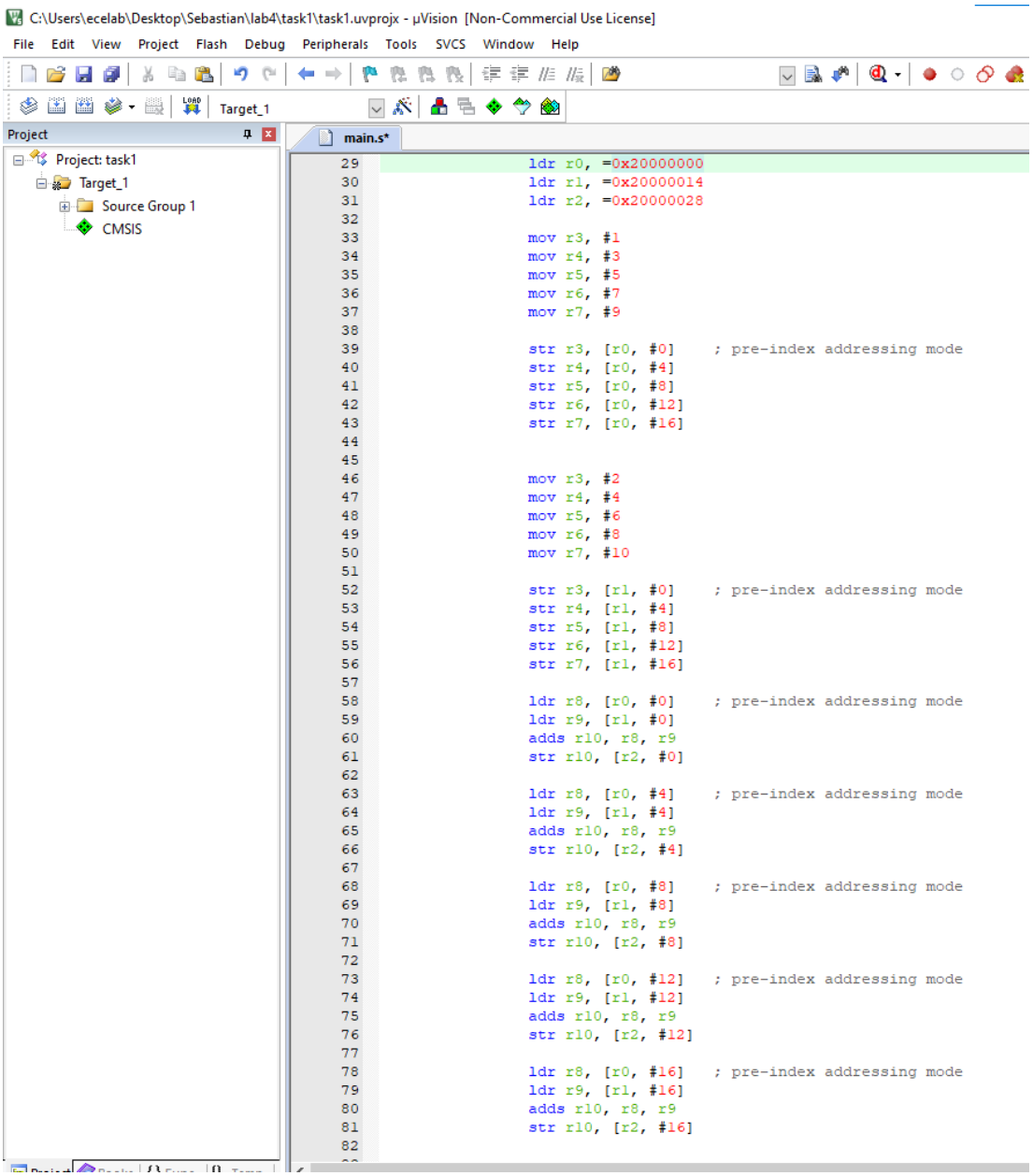
Task-1 – 2 points

1. Screenshot of the complete code.




```
ldr r0, =0x20000000
ldr r1, =0x20000014
ldr r2, =0x20000028

mov r3, #1
mov r4, #3
mov r5, #5
mov r6, #7
mov r7, #9

str r3, [r0, #0]     ; pre-index addressing mode
str r4, [r0, #4]
str r5, [r0, #8]
str r6, [r0, #12]
str r7, [r0, #16]


mov r3, #2
mov r4, #4
mov r5, #6
mov r6, #8
mov r7, #10

str r3, [r1, #0]     ; pre-index addressing mode
str r4, [r1, #4]
str r5, [r1, #8]
str r6, [r1, #12]
str r7, [r1, #16]

ldr r8, [r0, #0]     ; pre-index addressing mode
ldr r9, [r1, #0]
adds r10, r8, r9
str r10, [r2, #0]

ldr r8, [r0, #4]     ; pre-index addressing mode
ldr r9, [r1, #4]
adds r10, r8, r9
str r10, [r2, #4]

ldr r8, [r0, #8]     ; pre-index addressing mode
ldr r9, [r1, #8]
adds r10, r8, r9
str r10, [r2, #8]

ldr r8, [r0, #12]    ; pre-index addressing mode
ldr r9, [r1, #12]
adds r10, r8, r9
str r10, [r2, #12]

ldr r8, [r0, #16]    ; pre-index addressing mode
ldr r9, [r1, #16]
adds r10, r8, r9
str r10, [r2, #16]
```

2. Include the correct array initialization and addition

C:\Users\ecelab\Desktop\Sebastian\lab4\task1\task1.uvprojx - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Core	
R0	0x20000000
R1	0x20000014
R2	0x20000028
R3	0x00000001
R4	0x00000003
R5	0x00000005
R6	0x00000007
R7	0x00000009
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x080000F0
xPSR	0x01000000
N	0
Z	0
C	0
V	0
Q	0
GE	0x0
T	1
IT	Disabled
ISR	0
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	23
Sec	0.00000192
FPU	

Project | Registers

Disassembly

```
    43:                                 str r7, [r0, #16]
    44:
    45:
0x080000EE 6107      STR            r7,[r0,#0x10]
    46:                                 mov r3, #2
0x080000F0 F04F0302  MOV            r3,#0x02
```

main.s*

```
37                  mov r7, #9
38
39                  str r3, [r0, #0]     ; pre-index addressing mode
40                  str r4, [r0, #4]
41                  str r5, [r0, #8]
42                  str r6, [r0, #12]
43                  str r7, [r0, #16]
44
45
46                  mov r3, #2
47                  mov r4, #4
48                  mov r5, #6
49                  mov r6, #8
50                  mov r7, #10
51
52                  str r3, [r1, #0]     ; pre-index addressing mode
53                  str r4, [r1, #4]
54                  str r5, [r1, #8]
55                  str r6, [r1, #12]
56                  str r7, [r1, #16]
57
58                  ldr r8, [r0, #0]     ; pre-index addressing mode
59                  ldr r9, [r1, #0]
60                  adds r10, r8, r9
61                  str r10, [r2, #0]
62
63                  ldr r8, [r0, #4]     ; pre-index addressing mode
64                  ldr r9, [r1, #4]
65                  adds r10, r8, r9
66                  str r10, [r2, #4]
```

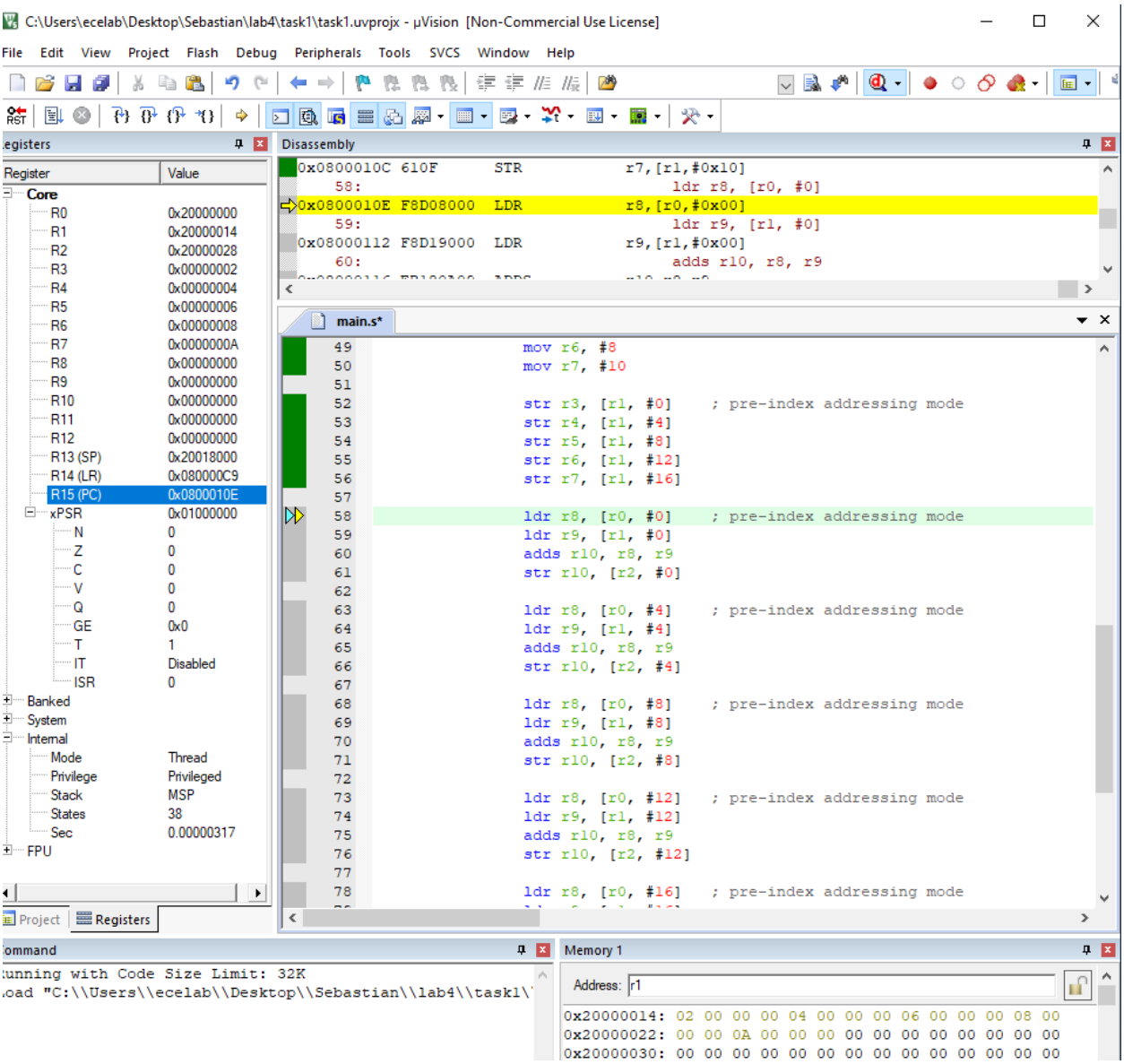
Command

```
Running with Code Size Limit: 32K
Load "C:\\Users\\ecelab\\Desktop\\Sebastian\\lab4\\task1\
```

Memory 1

Address: r0

```
0x20000000: 01 00 00 00 03 00 00 00 05 00 00 00 07 00
0x2000000E: 00 00 09 00 00 00 00 00 00 00 00 00 00 00
0x2000001C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```



3. Include the correct addressing mode used and the comments mentioned in the code
4. NVZC Flags screenshots for each step, whenever there is a change

R4	0x00000004
R5	0x00000006
R6	0x00000008
R7	0x0000000A
R8	0x00000001
R9	0x00000002
R10	0x00000003
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x0800011A
xPSR	0x01000000
N	0
Z	0
C	0
V	0
Q	0

main.s*

```
52  str r3, [r1, #0]    ; pre-index addressing mode
53  str r4, [r1, #4]
54  str r5, [r1, #8]
55  str r6, [r1, #12]
56  str r7, [r1, #16]
57
58  ldr r8, [r0, #0]    ; pre-index addressing mode
59  ldr r9, [r1, #0]
60  adds r10, r8, r9
61  str r10, [r2, #0]
62
63  ldr r8, [r0, #4]    ; pre-index addressing mode
64  ldr r9, [r1, #4]
65  adds r10, r8, r9
66  str r10, [r2, #4]
```

R9	0x00000004
R10	0x00000007
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x0800012A
xPSR	0x01000000
N	0
Z	0
C	0
V	0
Q	0
GE	0x0
T	1
IT	Disabled
ISR	0

```
59  ldr r9, [r1, #0]
60  adds r10, r8, r9
61  str r10, [r2, #0]
62
63  ldr r8, [r0, #4]    ; pre-index addressing mode
64  ldr r9, [r1, #4]
65  adds r10, r8, r9
66  str r10, [r2, #4]
67
68  ldr r8, [r0, #8]    ; pre-index addressing mode
69  ldr r9, [r1, #8]
70  adds r10, r8, r9
71  str r10, [r2, #8]
72
73  ldr r8, [r0, #12]   ; pre-index addressing mode
74  ldr r9, [r1, #12]
75  adds r10, r8, r9
```

R6	0x00000008
R7	0x0000000A
R8	0x00000005
R9	0x00000006
R10	0x0000000B
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x0800013E
xPSR	0x01000000
N	0
Z	0
C	0
V	0
Q	0
GE	0x0
T	1
IT	Disabled
ISR	0
Banked	
System	

```
64  ldr r9, [r1, #4]
65  adds r10, r8, r9
66  str r10, [r2, #4]
67
68  ldr r8, [r0, #8]    ; pre-index addressing mode
69  ldr r9, [r1, #8]
70  adds r10, r8, r9
71  str r10, [r2, #8]
72
73  ldr r8, [r0, #12]   ; pre-index addressing mode
74  ldr r9, [r1, #12]
75  adds r10, r8, r9
76  str r10, [r2, #12]
77
78  ldr r8, [r0, #16]   ; pre-index addressing mode
79  ldr r9, [r1, #16]
80  adds r10, r8, r9
81  str r10, [r2, #16]
82
83
```

R7	0x0000000A
R8	0x00000007
R9	0x00000008
R10	0x0000000F
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x0800014A
xPSR	0x01000000
N	0
Z	0
C	0
V	0
Q	0
GE	0x0
T	1
IT	Disabled
ISR	0
Banked	
System	

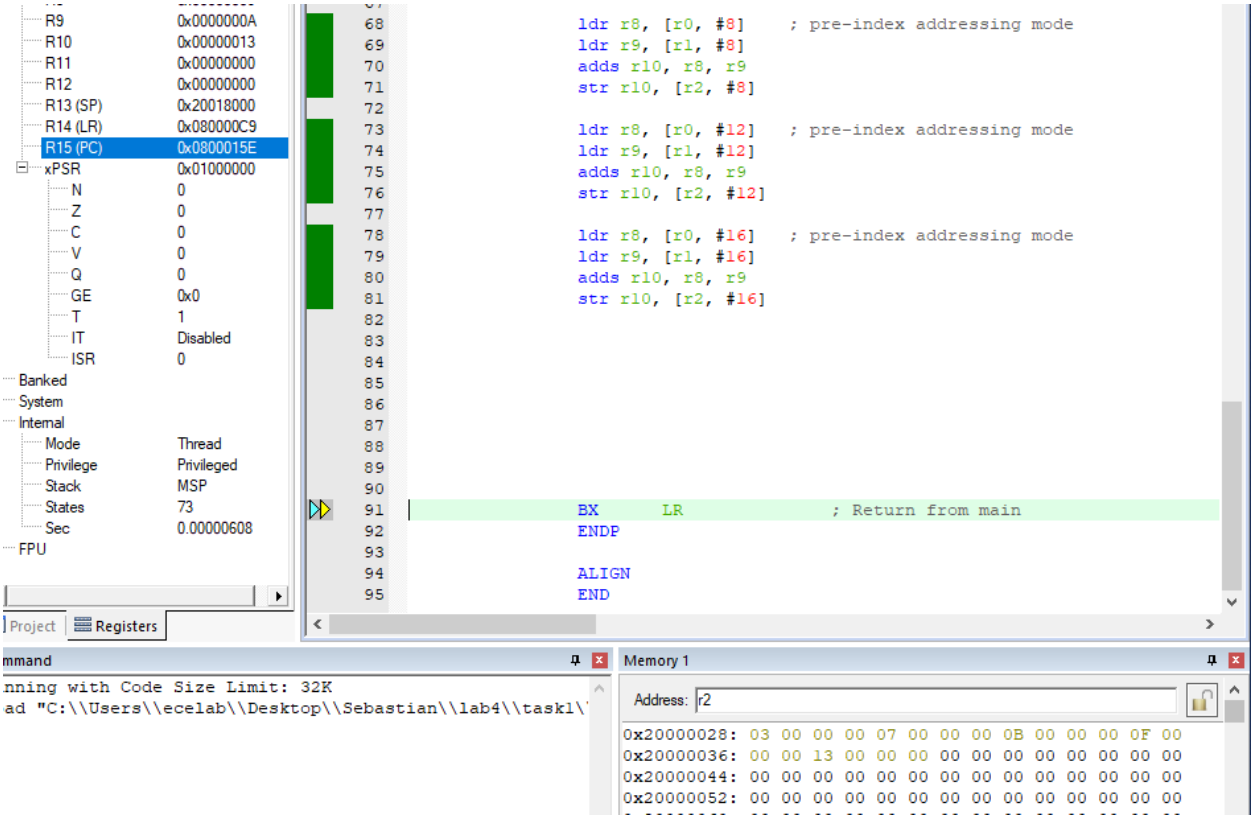
```
66          str r10, [r2, #4]
67
68          ldr r8, [r0, #8]     ; pre-index addressing mode
69          ldr r9, [r1, #8]
70          adds r10, r8, r9
71          str r10, [r2, #8]
72
73          ldr r8, [r0, #12]    ; pre-index addressing mode
74          ldr r9, [r1, #12]
75          adds r10, r8, r9
76          str r10, [r2, #12]
77
78          ldr r8, [r0, #16]    ; pre-index addressing mode
79          ldr r9, [r1, #16]
80          adds r10, r8, r9
81          str r10, [r2, #16]
82
83
84
85
86
```

R5	0x00000006
R6	0x00000008
R7	0x0000000A
R8	0x00000009
R9	0x0000000A
R10	0x00000013
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x0800015A
xPSR	0x01000000
N	0
Z	0
C	0
V	0
Q	0
GE	0x0
T	1
IT	Disabled
ISR	0
anked	
ystem	

main.s*

```
66          str r10, [r2, #4]
67
68          ldr r8, [r0, #8]     ; pre-index addressing mode
69          ldr r9, [r1, #8]
70          adds r10, r8, r9
71          str r10, [r2, #8]
72
73          ldr r8, [r0, #12]    ; pre-index addressing mode
74          ldr r9, [r1, #12]
75          adds r10, r8, r9
76          str r10, [r2, #12]
77
78          ldr r8, [r0, #16]    ; pre-index addressing mode
79          ldr r9, [r1, #16]
80          adds r10, r8, r9
81          str r10, [r2, #16]
82
83
84
85
86
```

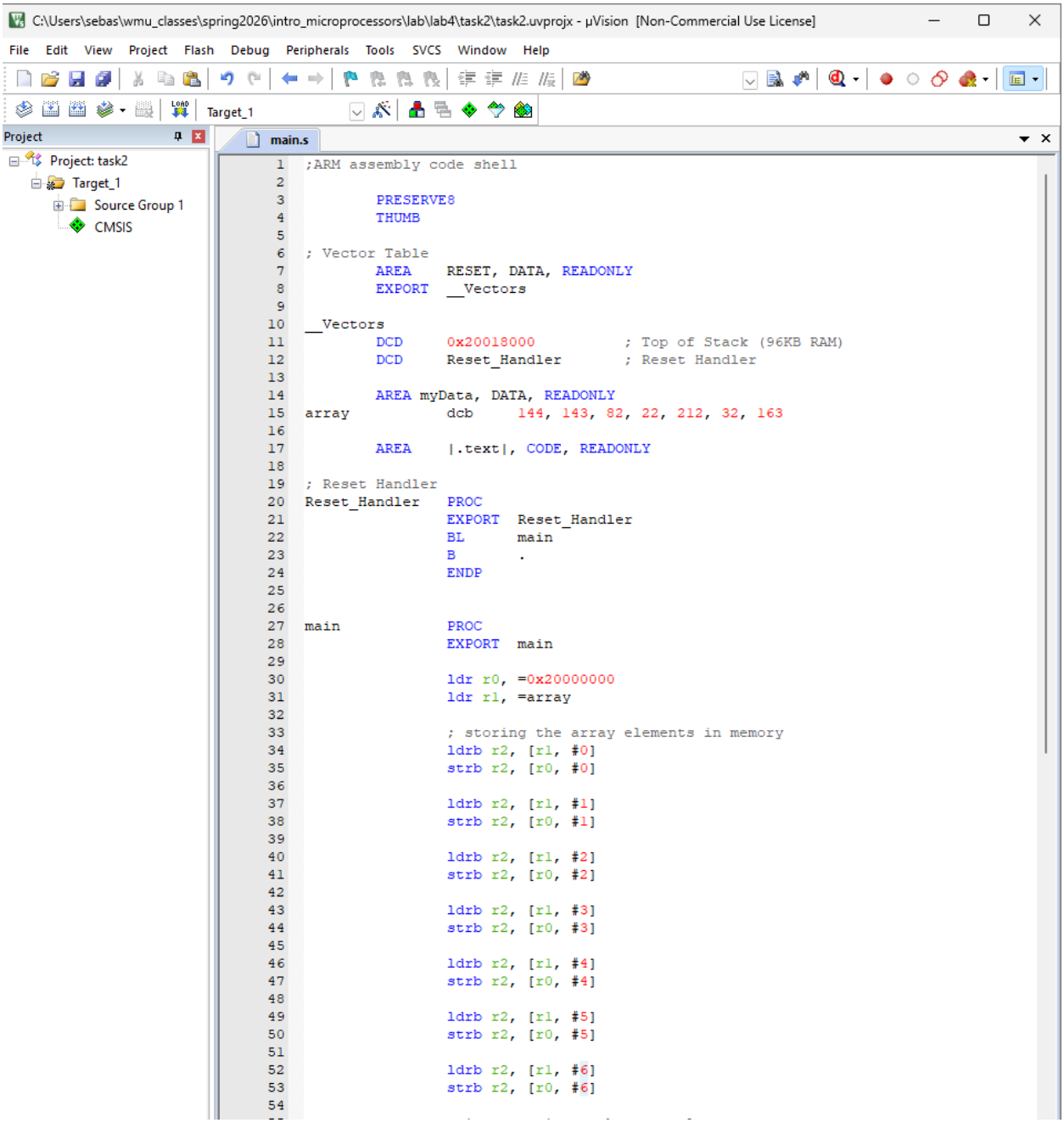
Array sum stored in 0x20000028



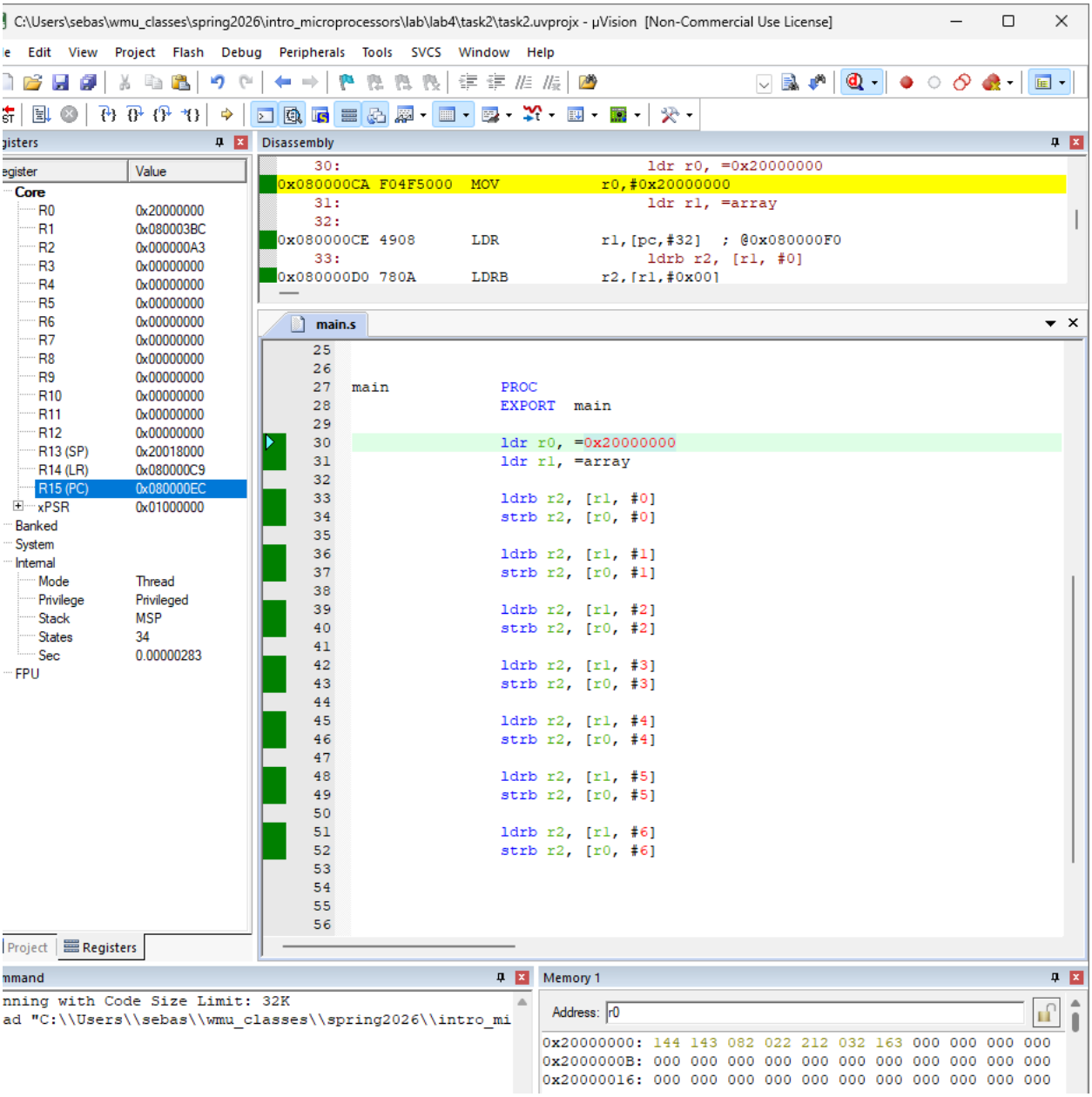
Task-2 – 2 points

1. Screenshot of the code for each addressing mode

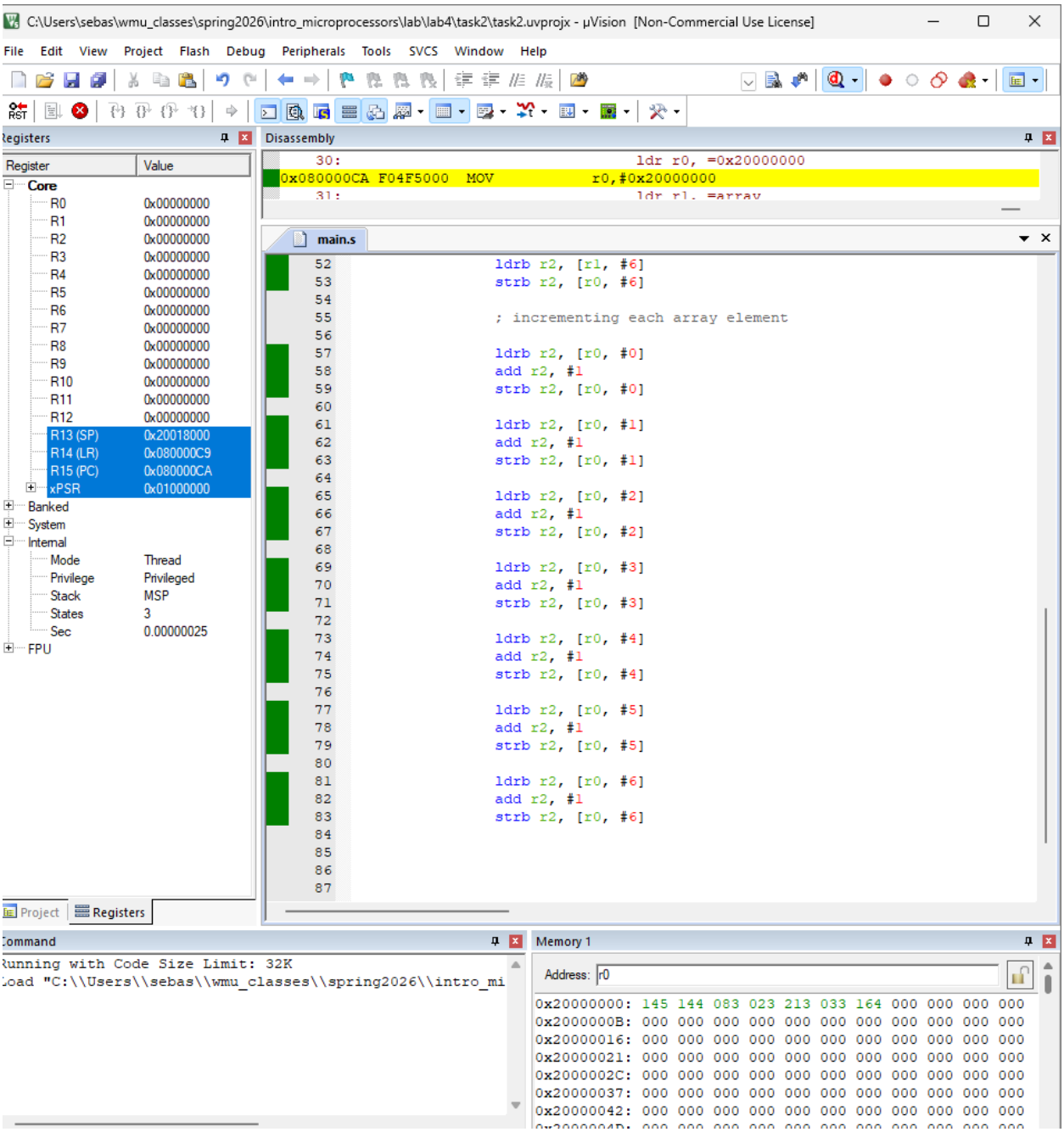
Storing array in memory address 0x20000000



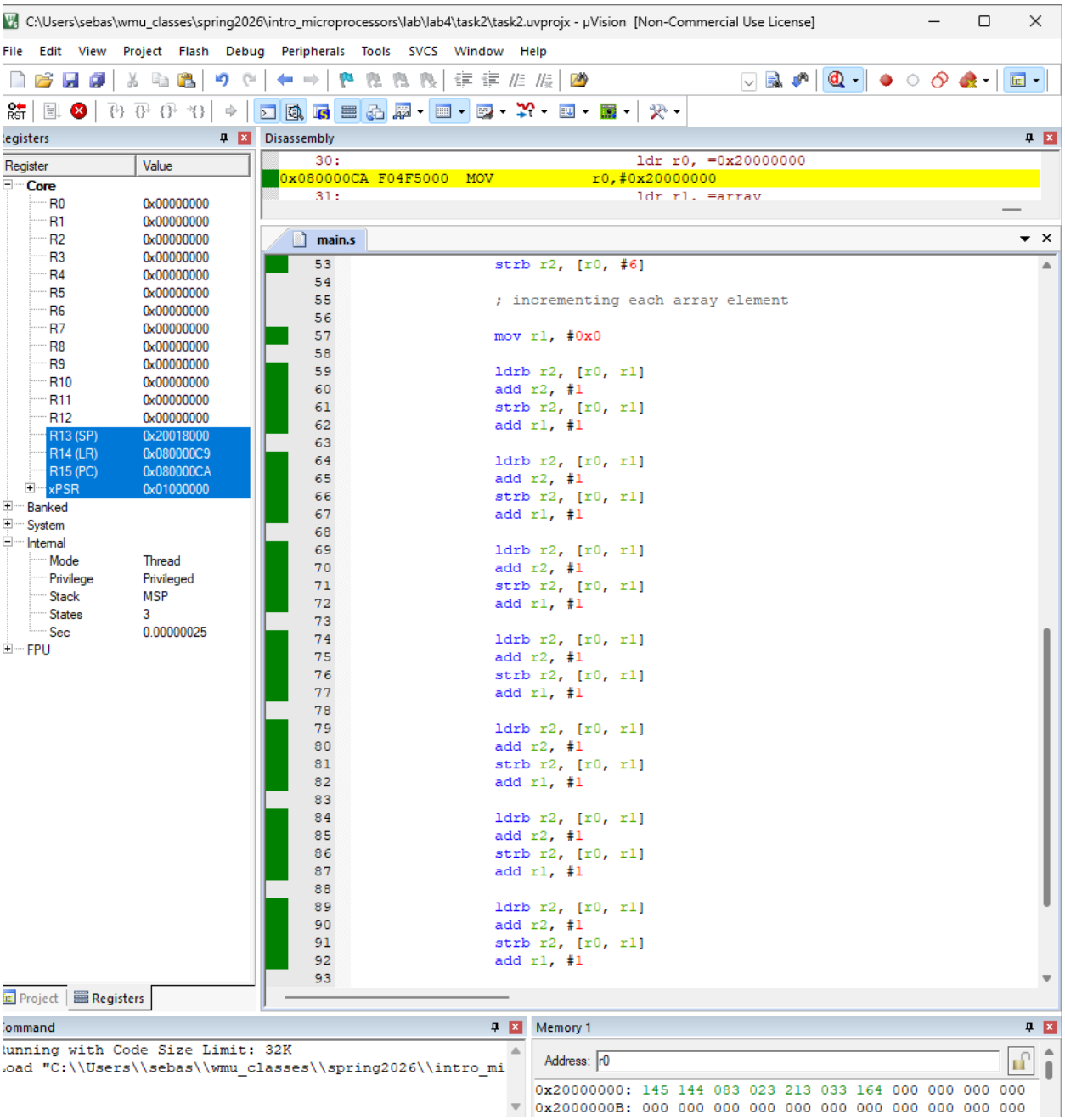

```
;ARM assembly code shell

        PRESERVE8
        THUMB

; Vector Table
        AREA    RESET, DATA, READONLY
        EXPORT  __Vectors

__Vectors
        DCD     0x20018000          ; Top of Stack (96KB RAM)
        DCD     Reset_Handler       ; Reset Handler

        AREA myData, DATA, READONLY
array           dcb     144, 143, 82, 22, 212, 32, 163

        AREA    |.text|, CODE, READONLY

; Reset Handler
Reset_Handler   PROC
                EXPORT  Reset_Handler
                BL      main
                B       .
                ENDP


main            PROC
                EXPORT  main

                ldr r0, =0x20000000
                ldr r1, =array

                ; storing the array elements in memory
                ldrb r2, [r1, #0]
                strb r2, [r0, #0]

                ldrb r2, [r1, #1]
                strb r2, [r0, #1]

                ldrb r2, [r1, #2]
                strb r2, [r0, #2]

                ldrb r2, [r1, #3]
                strb r2, [r0, #3]

                ldrb r2, [r1, #4]
                strb r2, [r0, #4]

                ldrb r2, [r1, #5]
                strb r2, [r0, #5]

                ldrb r2, [r1, #6]
                strb r2, [r0, #6]
```


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task2\task2.uvprojx - µVision [Non-Commercial Use License]
Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Disassembly
30: ldr r0, =0x20000000
0x080000CA F04F5000 MOV r0,#0x20000000
31: ldr r1, =array
32:
0x080000CE 4908 LDR r1,[pc,#32] ; @0x080000F0
33: ldrb r2, [r1, #0]
0x080000D0 780A LDRB r2,[r1,#0x00]
main.s
main PROC
EXPORT main
ldr r0, =0x20000000
ldr r1, =array
ldrb r2, [r1, #0]
strb r2, [r0, #0]
ldrb r2, [r1, #1]
strb r2, [r0, #1]
ldrb r2, [r1, #2]
strb r2, [r0, #2]
ldrb r2, [r1, #3]
strb r2, [r0, #3]
ldrb r2, [r1, #4]
strb r2, [r0, #4]
ldrb r2, [r1, #5]
strb r2, [r0, #5]
ldrb r2, [r1, #6]
strb r2, [r0, #6]
Register Value
Core
R0 0x20000000
R1 0x080003BC
R2 0x000000A3
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000EC
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 34
Sec 0.00000283
FPU
Project Registers
nning with Code Size Limit: 32K
ad "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: r0
0x20000000: 144 143 082 022 212 032 163 000 000 000 000
0x2000000B: 000 000 000 000 000 000 000 000 000 000 000
0x20000016: 000 000 000 000 000 000 000 000 000 000 000

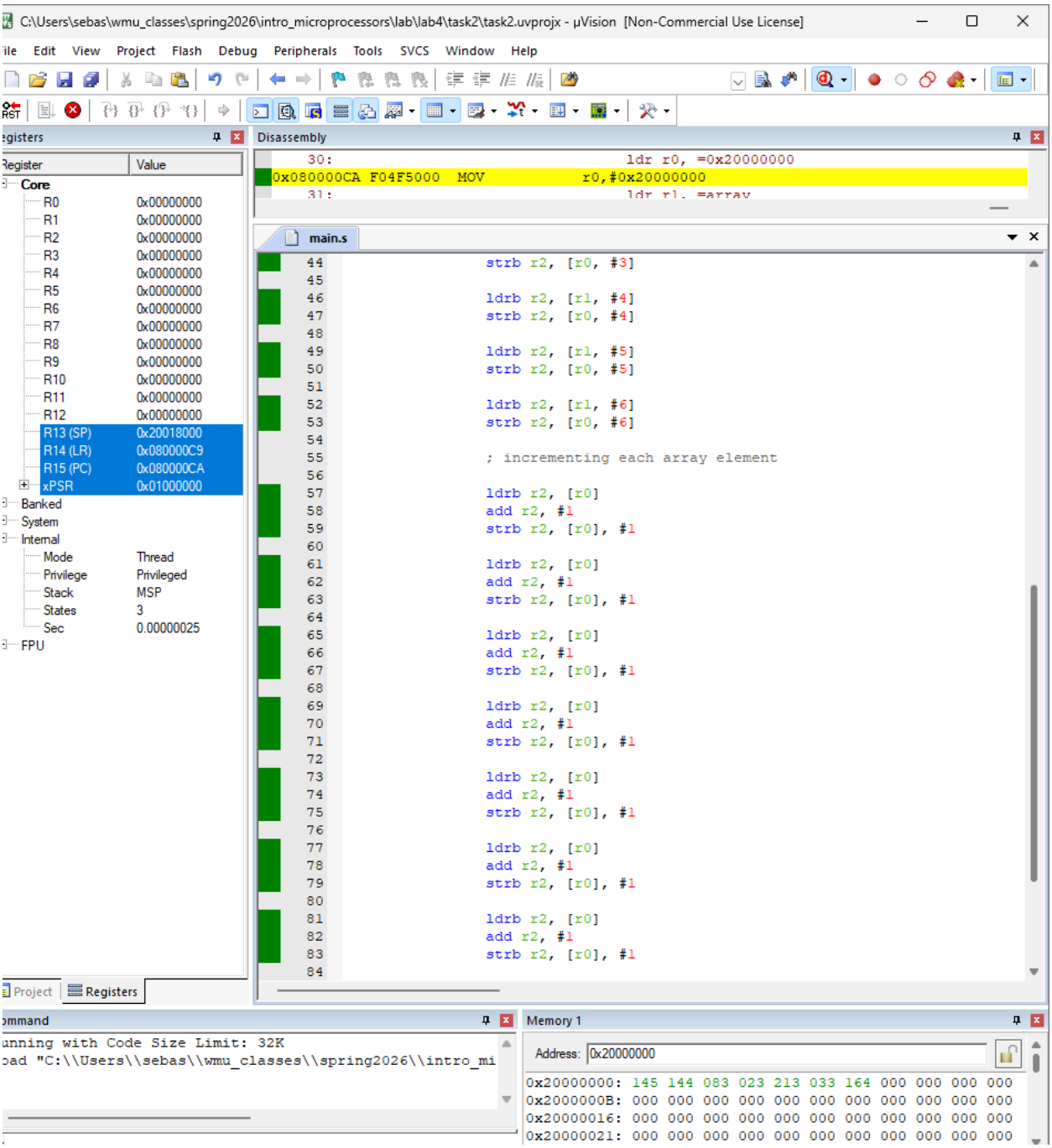
Pre indexing mode:



Post Indexing mode:



Post indexing:



2. RTL for post-index

R2 <- M[r0]

R2 <- R2 +1

M[r0] <- R2

R0 <- R0 +1

R2 <- M[r0]

R2 <- R2 +1

M[r0] <- R2

R0 <- R0 +1

R2 <- M[r0]

R2 <- R2 +1

M[r0] <- R2

R0 <- R0 +1

R2 <- M[r0]

R2 <- R2 +1

M[r0] <- R2

R0 <- R0 +1

M[r0] <- R2

R0 <- R0 +1

R2 <- M[r0]

R2 <- R2 +1

M[r0] <- R2

R0 <- R0 +1

R2 <- M[r0]

R2 <- R2 +1

M[r0] <- R2

R0 <- R0 +1

R2 <- M[r0]

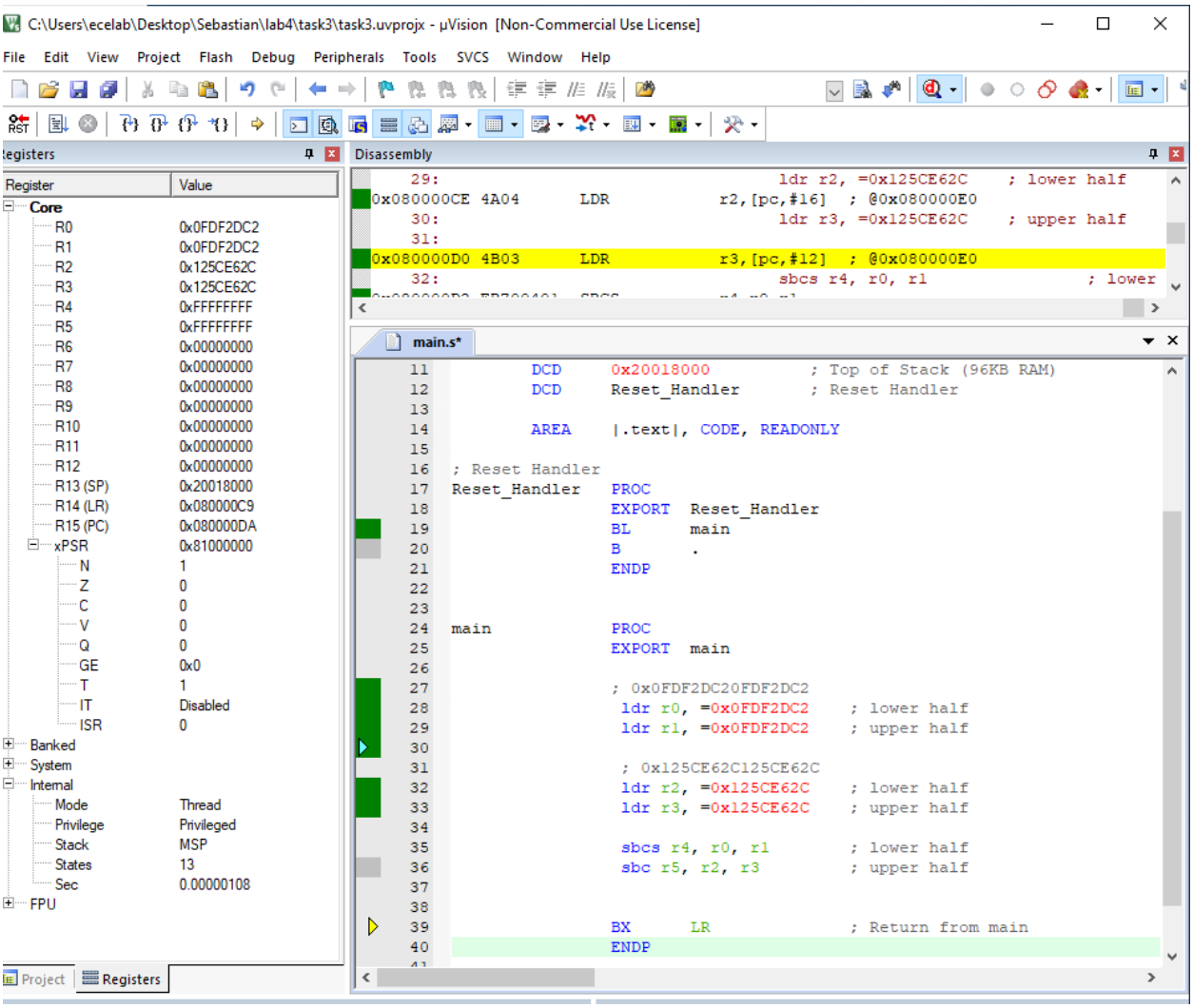
R2 <- R2 +1

M[r0] <- R2

R0 <- R0 +1

Task-3 – 2 points

1. Screenshot of the code.



2. Byte-by-byte subtraction with borrow handling result

C:\Users\ecelab\Desktop\Sebastian\lab4\task3\task3.uvprojx - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Target_1

roject

Project: task3
Target_1
Source Group 1
CMSIS

main.s

```
                ENDP


main            PROC
                EXPORT  main

                ldr r8, =0x20000000
                ldr r9, =0x20000014
                ldr r10, =0x20000028

                ; 0x0FDF2DC20FDF2DC2
                 ldr r0, =0x0FDF2DC2    ; lower half
                 ldr r1, =0x0FDF2DC2    ; upper half

                ; 0x125CE62C125CE62C
                 ldr r2, =0x125CE62C    ; lower half
                 ldr r3, =0x125CE62C    ; upper half

                 ;sbcs r4, r0, r1       ; lower half
                 ;sbc r5, r2, r3            ; upper half

                 ; storing first number's bytes

                 ldr r4, =0xC2
                 ldr r5, =0x2D
                 ldr r6, =0xDF
                 ldr r7, =0x0F

                 strb r4, [r8, #0]
                 strb r5, [r8, #1]
                 strb r6, [r8, #2]
                 strb r7, [r8, #3]

                 ; storing seecond number's bytes

                 ldr r4, =0x2C
                 ldr r5, =0xE6
                 ldr r6, =0x5C
                 ldr r7, =0x12

                 strb r4, [r9, #0]
                 strb r5, [r9, #1]
                 strb r6, [r9, #2]
                 strb r7, [r9, #3]
```

C:\Users\ecelab\Desktop\Sebastian\lab4\task3\task3.uvprojx - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Target_1

Project

- Project: task3
  - Target_1
    - Source Group 1
    - CMSIS

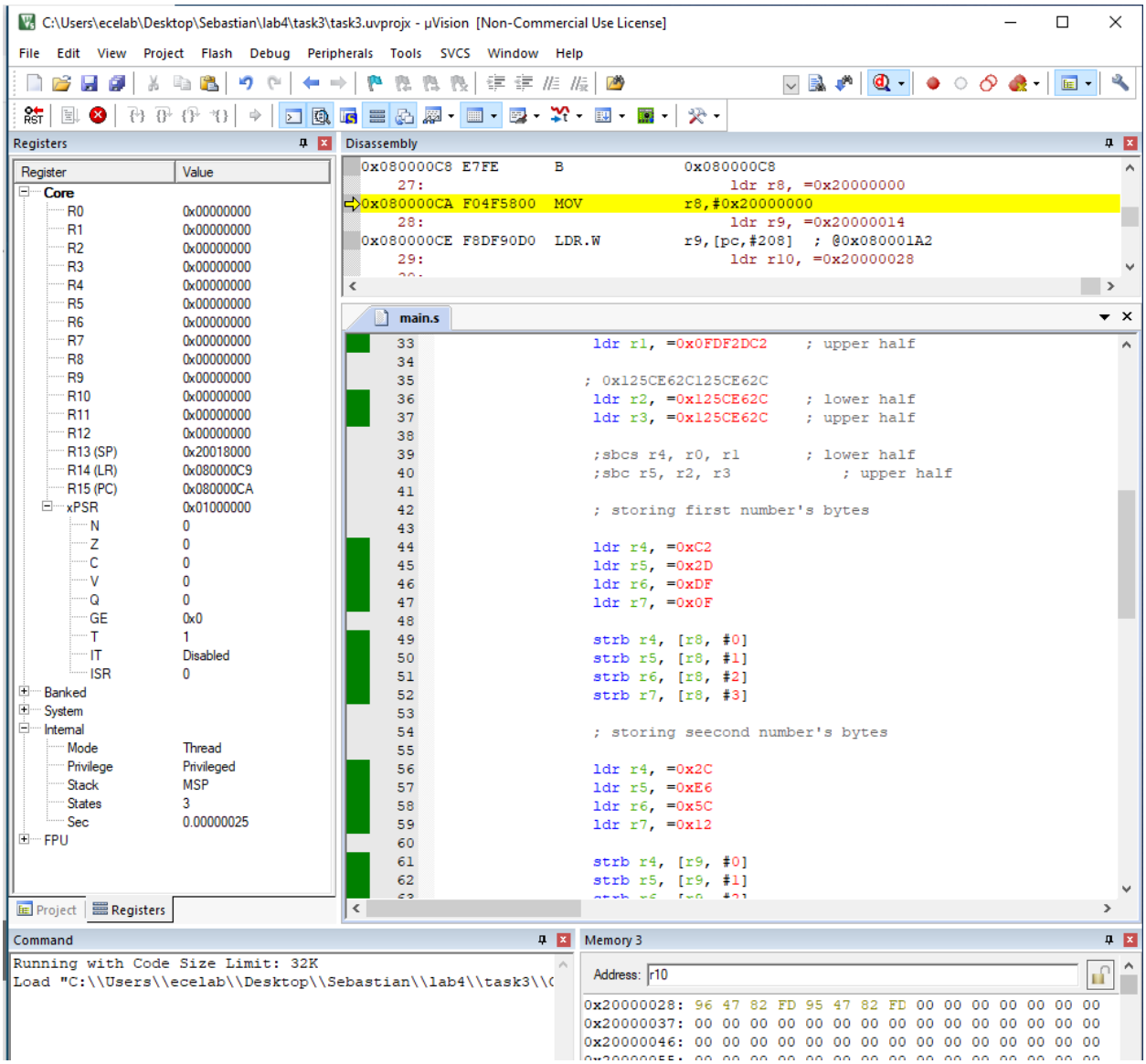
main.s

```
                strb r6, [r9, #2]
                strb r7, [r9, #3]

                ; substraction byte by byte
                ldrb r4, [r8, #0]       ; first byte
                ldrb r5, [r9, #0]

                subs r6, r4, r5
                strb r6, [r10, #0]

                ldrb r4, [r8, #1]       ; second byte
                ldrb r5, [r9, #1]

                sbcs r6, r4, r5
                strb r6, [r10, #1]

                ldrb r4, [r8, #2]       ; third byte
                ldrb r5, [r9, #2]

                sbcs r6, r4, r5
                strb r6, [r10, #2]

                ldrb r4, [r8, #3]       ; fourth byte
                ldrb r5, [r9, #3]

                sbcs r6, r4, r5
                strb r6, [r10, #3]

                ldrb r4, [r8, #0]       ; fifth byte
                ldrb r5, [r9, #0]

                sbcs r6, r4, r5
                strb r6, [r10, #4]

                ldrb r4, [r8, #1]       ; sixth byte
                ldrb r5, [r9, #1]

                sbcs r6, r4, r5
                strb r6, [r10, #5]

                ldrb r4, [r8, #2]       ; seventh byte
                ldrb r5, [r9, #2]

                sbcs r6, r4, r5
                strb r6, [r10, #6]

                ldrb r4, [r8, #3]       ; eighth byte
                ldrb r5, [r9, #3]

                sbcs r6, r4, r5
                strb r6, [r10, #7]
```

Select the target to use

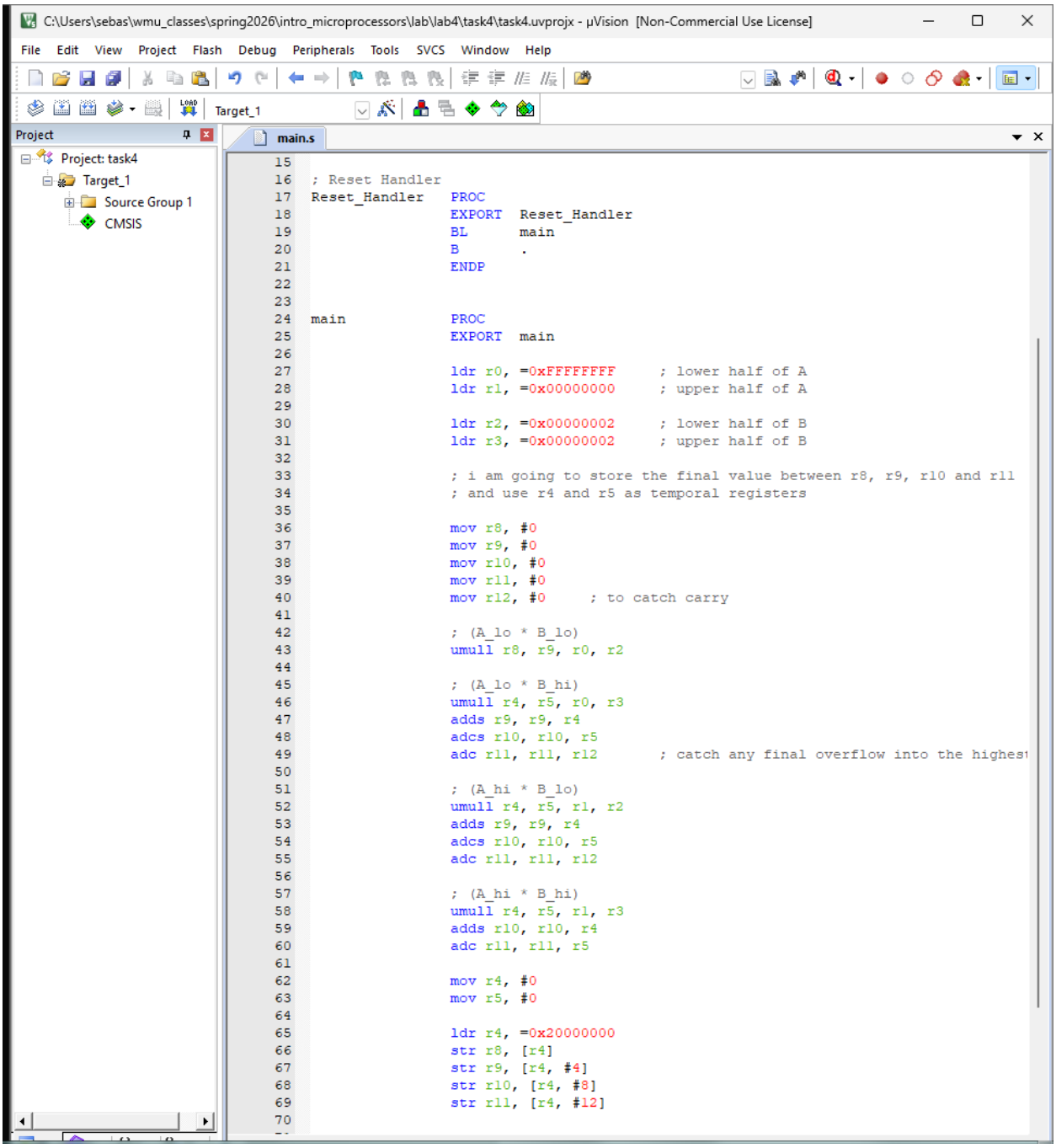
Simulation

3. Little-endian storage



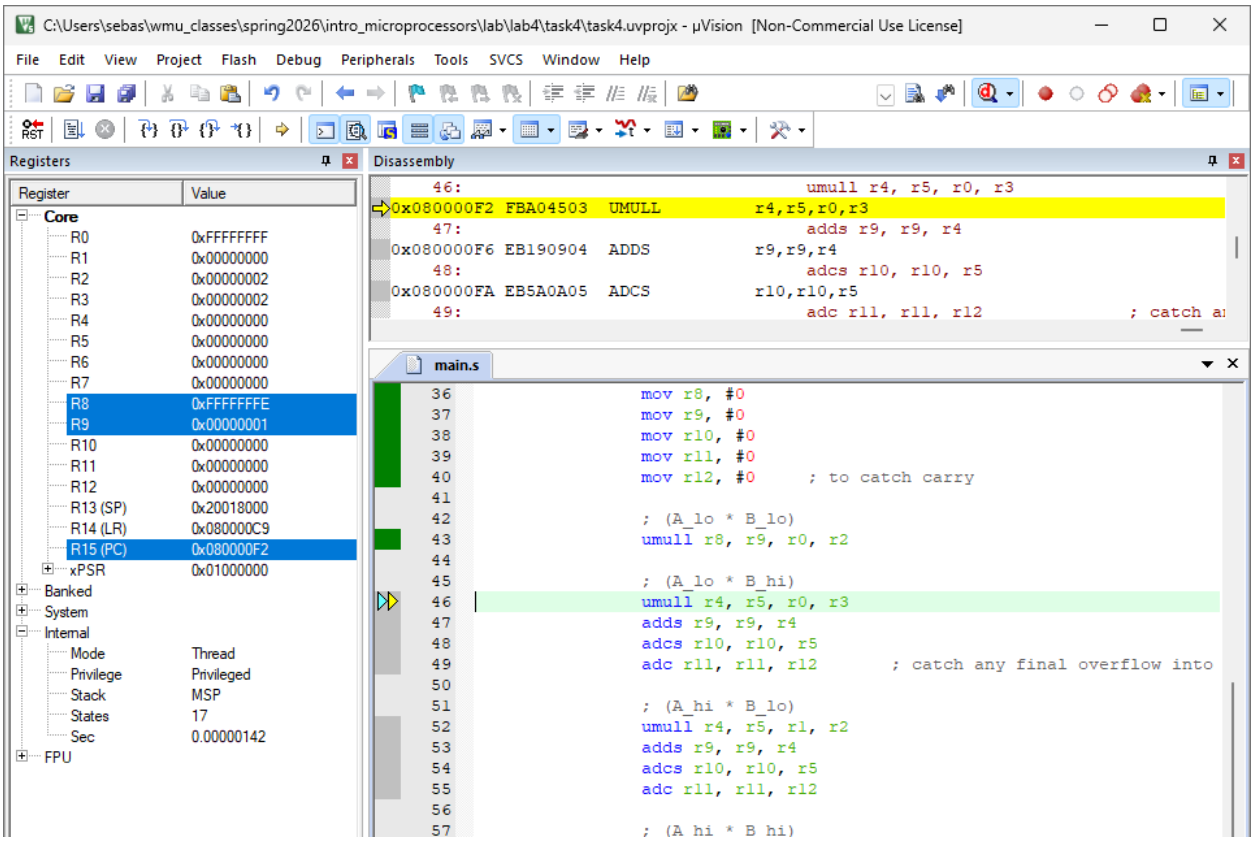
Task-4 – 2 points

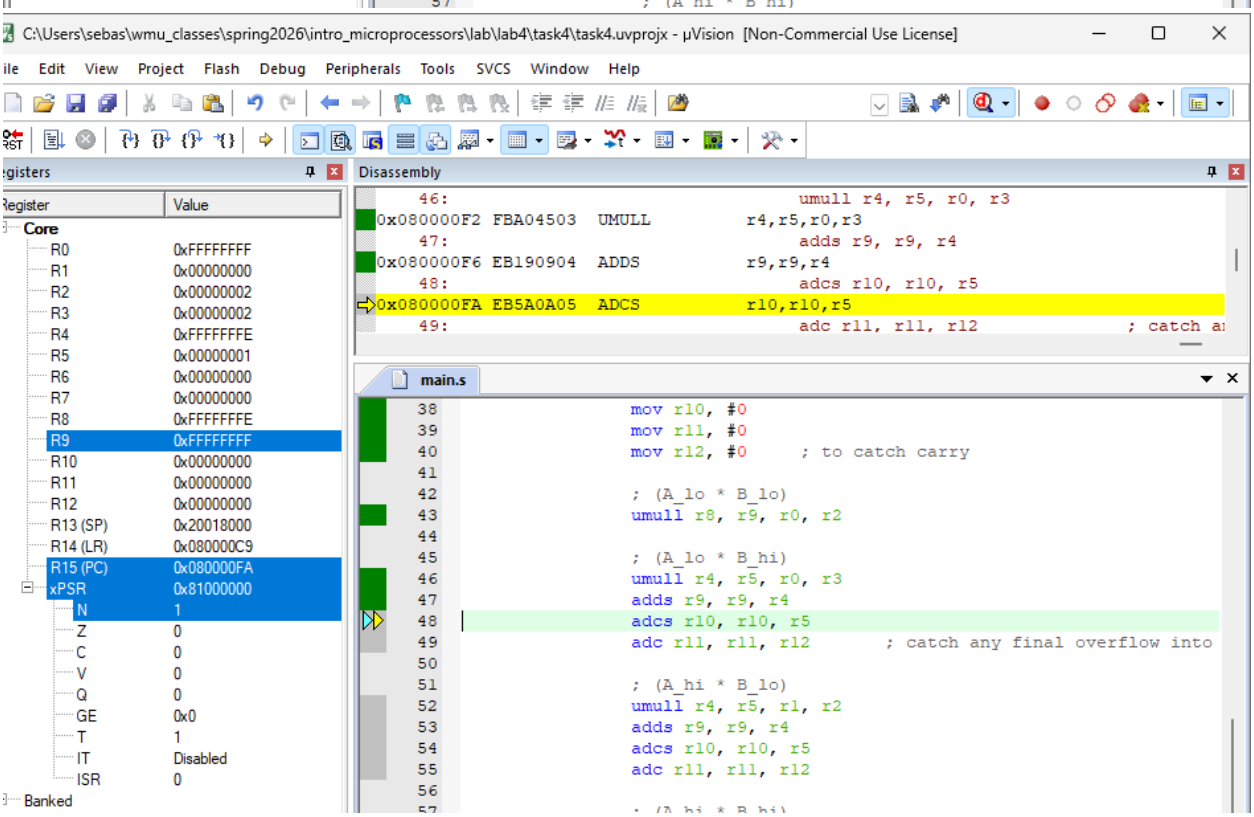
1. Screenshot of the code.




```
; Reset Handler
Reset_Handler   PROC
                EXPORT  Reset_Handler
                BL      main
                B       .
                ENDP


main            PROC
                EXPORT  main

                ldr r0, =0xFFFFFFFF     ; lower half of A
                ldr r1, =0x00000000     ; upper half of A

                ldr r2, =0x00000002     ; lower half of B
                ldr r3, =0x00000002     ; upper half of B

                ; i am going to store the final value between r8, r9, r10 and r11
                ; and use r4 and r5 as temporal registers

                mov r8, #0
                mov r9, #0
                mov r10, #0
                mov r11, #0
                mov r12, #0     ; to catch carry

                ; (A_lo * B_lo)
                umull r8, r9, r0, r2

                ; (A_lo * B_hi)
                umull r4, r5, r0, r3
                adds r9, r9, r4
                adcs r10, r10, r5
                adc r11, r11, r12       ; catch any final overflow into the highest

                ; (A_hi * B_lo)
                umull r4, r5, r1, r2
                adds r9, r9, r4
                adcs r10, r10, r5
                adc r11, r11, r12

                ; (A_hi * B_hi)
                umull r4, r5, r1, r3
                adds r10, r10, r4
                adc r11, r11, r5

                mov r4, #0
                mov r5, #0

                ldr r4, =0x20000000
                str r8, [r4]
                str r9, [r4, #4]
                str r10, [r4, #8]
                str r11, [r4, #12]
```

2. Correct Multiplication and division of 64-bit
3. NZCV flags screenshot for each step, whenever there is a change.


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task4\task4.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0xFFFFFFFF
R1 0x00000000
R2 0x00000002
R3 0x00000002
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0xFFFFFFFE
R9 0x00000001
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000F2
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 17
Sec 0.00000142
FPU
Disassembly
46: umull r4, r5, r0, r3
0x080000F2 FBA04503 UMULL r4,r5,r0,r3
47: adds r9, r9, r4
0x080000F6 EB190904 ADDS r9,r9,r4
48: adcs r10, r10, r5
0x080000FA EB5A0A05 ADCS r10,r10,r5
49: adc r11, r11, r12 ; catch a
main.s
36 mov r8, #0
37 mov r9, #0
38 mov r10, #0
39 mov r11, #0
40 mov r12, #0 ; to catch carry
41
42 ; (A_lo * B_lo)
43 umull r8, r9, r0, r2
44
45 ; (A_lo * B_hi)
46 umull r4, r5, r0, r3
47 adds r9, r9, r4
48 adcs r10, r10, r5
49 adc r11, r11, r12 ; catch any final overflow into
50
51 ; (A_hi * B_lo)
52 umull r4, r5, r1, r2
53 adds r9, r9, r4
54 adcs r10, r10, r5
55 adc r11, r11, r12
56
57 ; (A_hi * B_hi)


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task4\task4.uvprojx - µVision [Non-Commercial Use License]
ile Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Disassembly
46: umull r4, r5, r0, r3
0x080000F2 FBA04503 UMULL r4,r5,r0,r3
47: adds r9, r9, r4
0x080000F6 EB190904 ADDS r9,r9,r4
48: adcs r10, r10, r5
0x080000FA EB5A0A05 ADCS r10,r10,r5
49: adc r11, r11, r12 ; catch a
main.s
38 mov r10, #0
39 mov r11, #0
40 mov r12, #0 ; to catch carry
41
42 ; (A_lo * B_lo)
43 umull r8, r9, r0, r2
44
45 ; (A_lo * B_hi)
46 umull r4, r5, r0, r3
47 adds r9, r9, r4
48 adcs r10, r10, r5
49 adc r11, r11, r12 ; catch any final overflow into
50
51 ; (A_hi * B_lo)
52 umull r4, r5, r1, r2
53 adds r9, r9, r4
54 adcs r10, r10, r5
55 adc r11, r11, r12
56

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task4\task4.uvprojx - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Core	
R0	0xFFFFFFFF
R1	0x00000000
R2	0x00000002
R3	0x00000002
R4	0xFFFFFFFE
R5	0x00000001
R6	0x00000000
R7	0x00000000
R8	0xFFFFFFFE
R9	0xFFFFFFFF
R10	0x00000001
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x080000FE
xPSR	0x01000000
N	0
Z	0
C	0
V	0
Q	0
GE	0x0

Disassembly

```
0x080000FA EB5A0A05  ADCS          r10,r10,r5
    49:                                adc r11, r11, r12              ; catch a
    50:
    51:                                ; (A_hi * B_lo)
0x080000FE EB4B0B0C  ADC           r11,r11,r12
    52:                                umull r4, r5, r1, r2
0x08000102 FBA14502  UMULL         r4,r5,r1,r2
```

main.s

```
39                  mov r11, #0
40                  mov r12, #0      ; to catch carry
41
42                  ; (A_lo * B_lo)
43                  umull r8, r9, r0, r2
44
45                  ; (A_lo * B_hi)
46                  umull r4, r5, r0, r3
47                  adds r9, r9, r4
48                  adcs r10, r10, r5
49                  adc r11, r11, r12        ; catch any final overflow into
50
51                  ; (A_hi * B_lo)
52                  umull r4, r5, r1, r2
53                  adds r9, r9, r4
```

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task4\task4.uvprojx - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

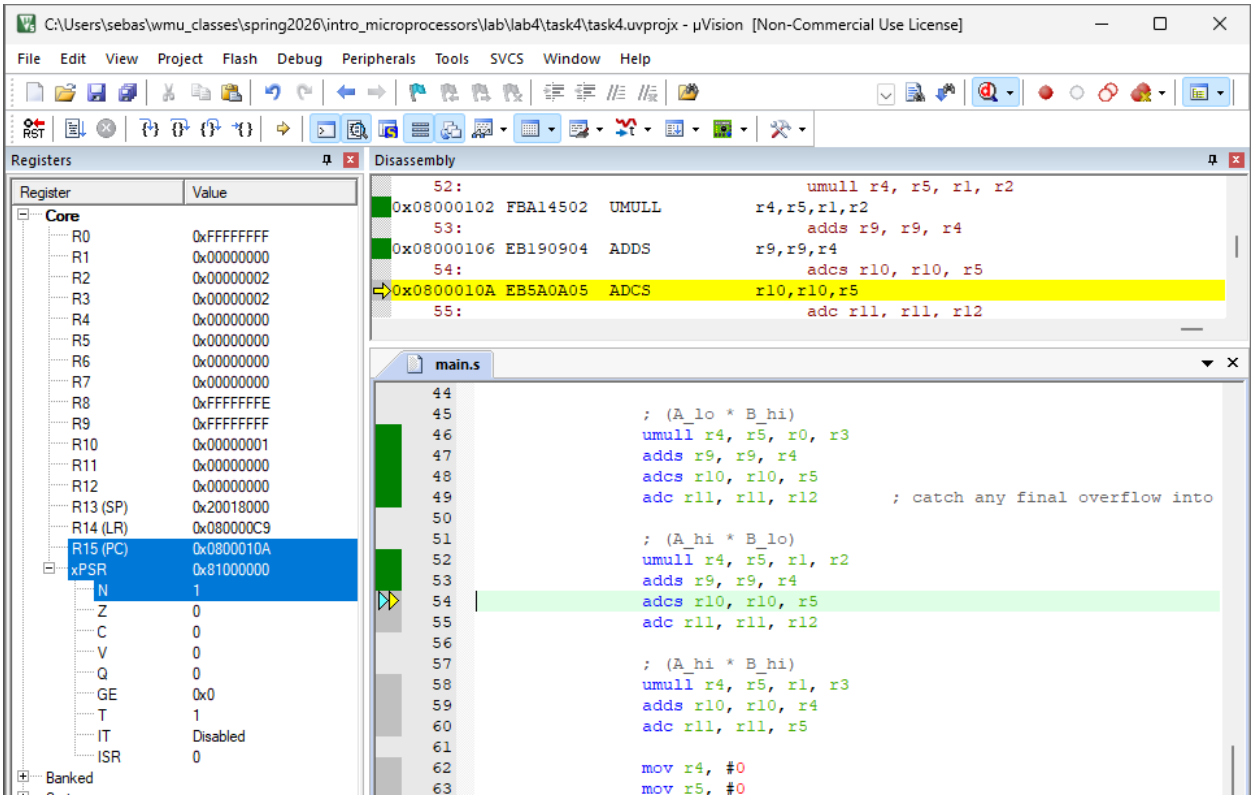
Register	Value
Core	
R0	0xFFFFFFFF
R1	0x00000000
R2	0x00000002
R3	0x00000002
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0xFFFFFFFE
R9	0xFFFFFFFF
R10	0x00000001
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x08000106
xPSR	0x01000000
N	0
Z	0
C	0
V	0
Q	0
GE	0x0
T	1
IT	Disabled
ISR	0

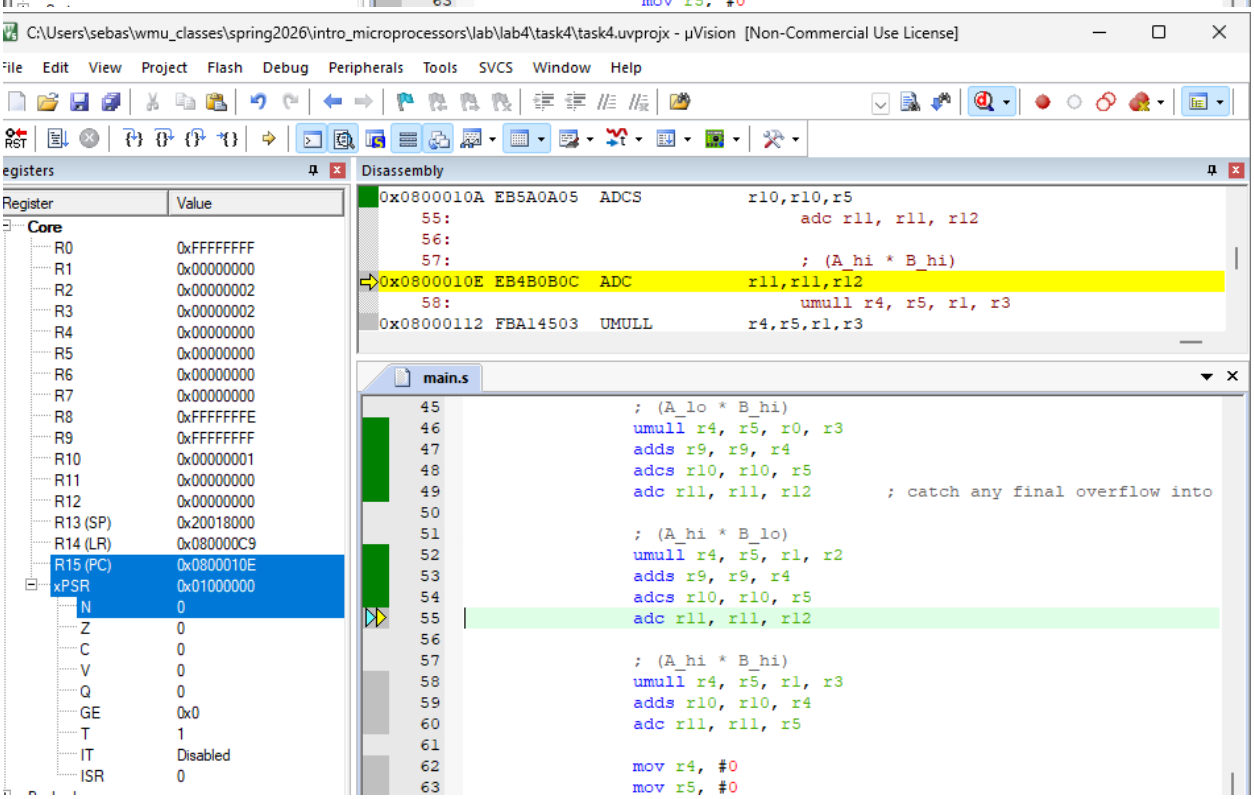
Disassembly

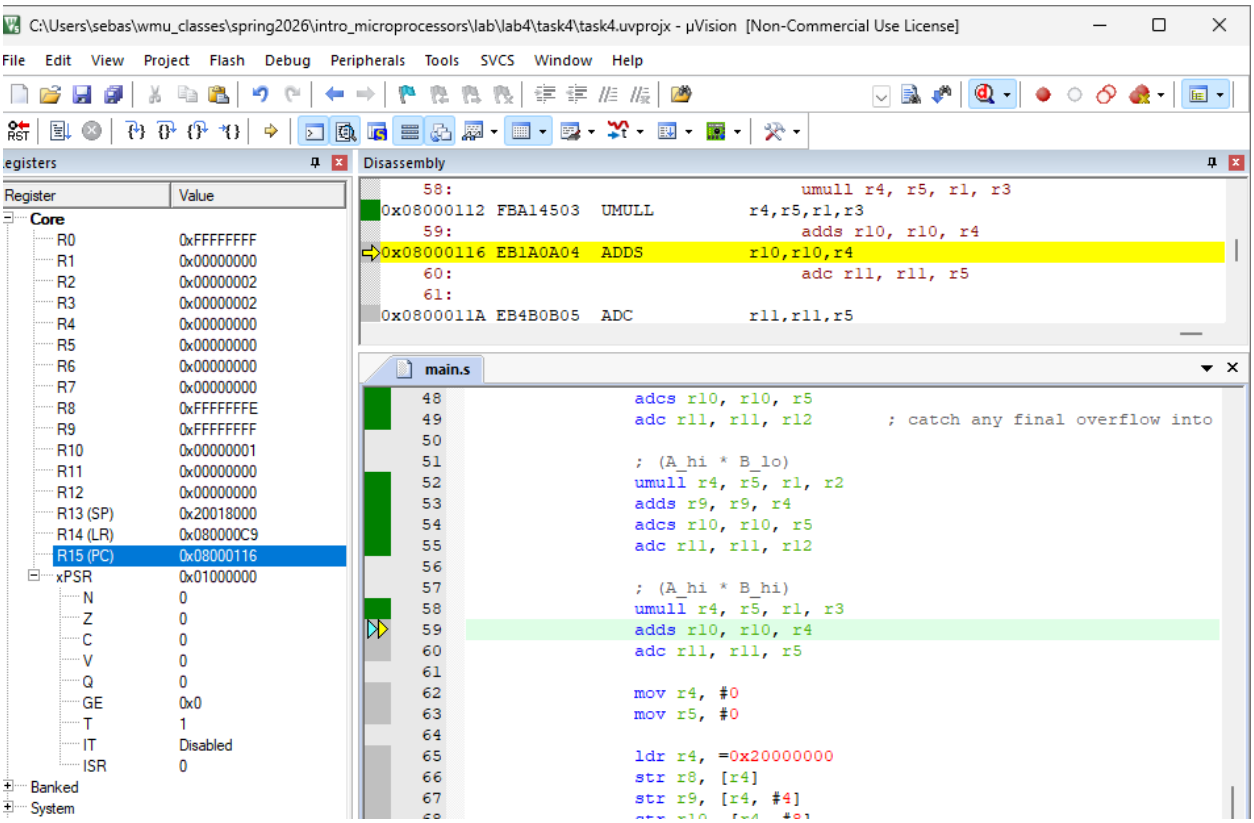
```
    52:                                umull r4, r5, r1, r2
0x08000102 FBA14502  UMULL         r4,r5,r1,r2
    53:                                adds r9, r9, r4
0x08000106 EB190904  ADDS          r9,r9,r4
    54:                                adcs r10, r10, r5
0x0800010A EB5A0A05  ADCS          r10,r10,r5
    55:                                adc r11, r11, r12
```

main.s

```
43                  umull r8, r9, r0, r2
44
45                  ; (A_lo * B_hi)
46                  umull r4, r5, r0, r3
47                  adds r9, r9, r4
48                  adcs r10, r10, r5
49                  adc r11, r11, r12        ; catch any final overflow into
50
51                  ; (A_hi * B_lo)
52                  umull r4, r5, r1, r2
53                  adds r9, r9, r4
54                  adcs r10, r10, r5
55                  adc r11, r11, r12
56
57                  ; (A_hi * B_hi)
58                  umull r4, r5, r1, r3
59                  adds r10, r10, r4
60                  adc r11, r11, r5
```

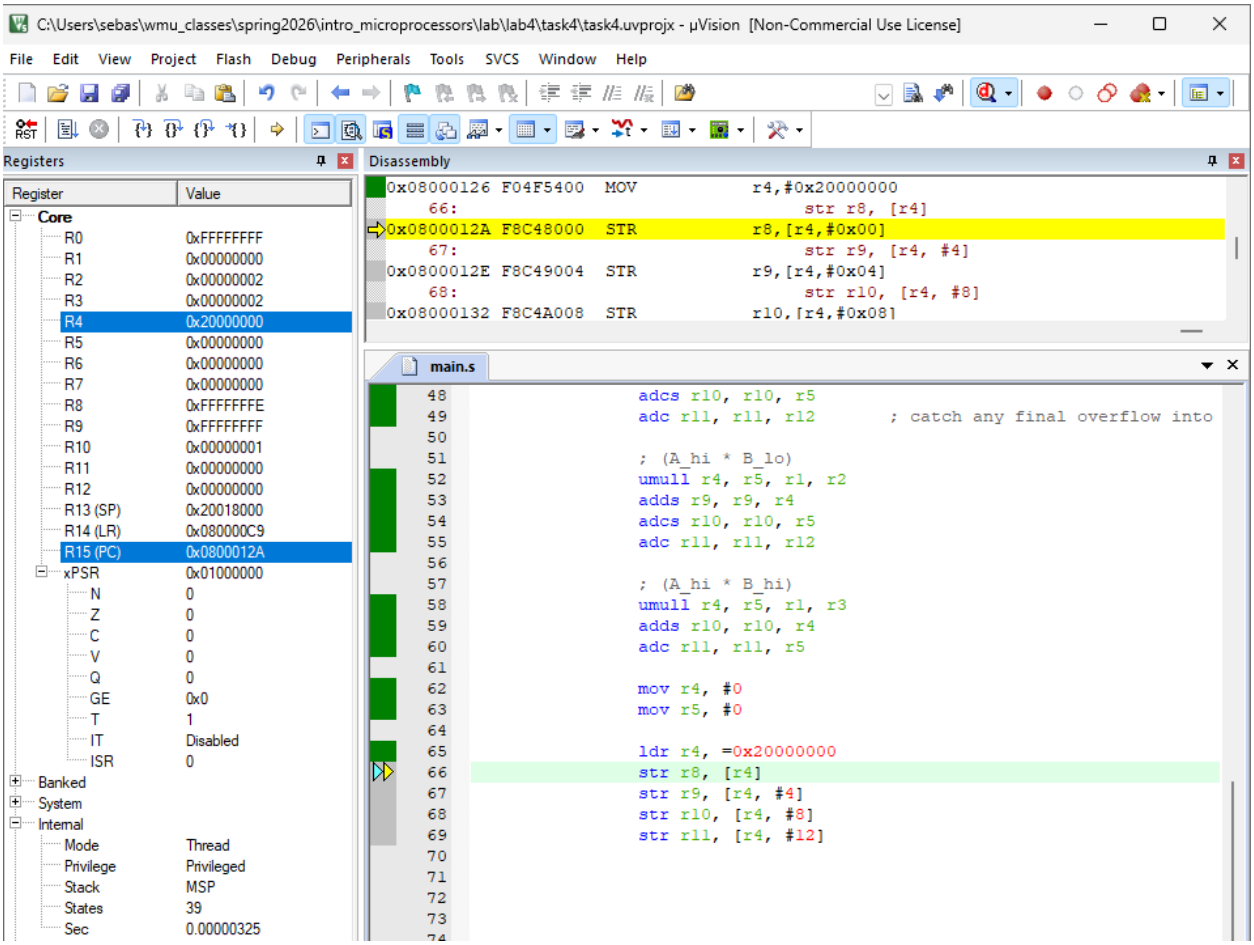

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task4\task4.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0xFFFFFFFF
R1 0x00000000
R2 0x00000002
R3 0x00000002
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0xFFFFFFFE
R9 0xFFFFFFFF
R10 0x00000001
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x0800010A
xPSR 0x81000000
N 1
Z 0
C 0
V 0
Q 0
GE 0x0
T 1
IT Disabled
ISR 0
Banked
Disassembly
52: umull r4, r5, r1, r2
0x08000102 FBA14502 UMULL r4,r5,r1,r2
53: adds r9, r9, r4
0x08000106 EB190904 ADDS r9,r9,r4
54: adcs r10, r10, r5
0x0800010A EB5A0A05 ADCS r10,r10,r5
55: adc r11, r11, r12
main.s
44
45 ; (A_lo * B_hi)
46 umull r4, r5, r0, r3
47 adds r9, r9, r4
48 adcs r10, r10, r5
49 adc r11, r11, r12 ; catch any final overflow into
50
51 ; (A_hi * B_lo)
52 umull r4, r5, r1, r2
53 adds r9, r9, r4
54 adcs r10, r10, r5
55 adc r11, r11, r12
56
57 ; (A_hi * B_hi)
58 umull r4, r5, r1, r3
59 adds r10, r10, r4
60 adc r11, r11, r5
61
62 mov r4, #0
63 mov r5, #0


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task4\task4.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0xFFFFFFFF
R1 0x00000000
R2 0x00000002
R3 0x00000002
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0xFFFFFFFE
R9 0xFFFFFFFF
R10 0x00000001
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x0800010E
xPSR 0x01000000
N 0
Z 0
C 0
V 0
Q 0
GE 0x0
T 1
IT Disabled
ISR 0
Disassembly
0x0800010A EB5A0A05 ADCS r10,r10,r5
55: adc r11, r11, r12
56:
57: ; (A_hi * B_hi)
0x0800010E EB4B0B0C ADC r11,r11,r12
58: umull r4, r5, r1, r3
0x08000112 FBA14503 UMULL r4,r5,r1,r3
main.s
45 ; (A_lo * B_hi)
46 umull r4, r5, r0, r3
47 adds r9, r9, r4
48 adcs r10, r10, r5
49 adc r11, r11, r12 ; catch any final overflow into
50
51 ; (A_hi * B_lo)
52 umull r4, r5, r1, r2
53 adds r9, r9, r4
54 adcs r10, r10, r5
55 adc r11, r11, r12
56
57 ; (A_hi * B_hi)
58 umull r4, r5, r1, r3
59 adds r10, r10, r4
60 adc r11, r11, r5
61
62 mov r4, #0
63 mov r5, #0



No more changes

Storing the final sum


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task4\task4.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0xFFFFFFFF
R1 0x00000000
R2 0x00000002
R3 0x00000002
R4 0x20000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0xFFFFFFFE
R9 0xFFFFFFFF
R10 0x00000001
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x0800012A
xPSR 0x01000000
N 0
Z 0
C 0
V 0
Q 0
GE 0x0
T 1
IT Disabled
ISR 0
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 39
Sec 0.00000325
Disassembly
0x08000126 F04F5400 MOV r4,#0x20000000
66: str r8, [r4]
0x0800012A F8C48000 STR r8,[r4,#0x00]
67: str r9, [r4, #4]
0x0800012E F8C49004 STR r9,[r4,#0x04]
68: str r10, [r4, #8]
0x08000132 F8C4A008 STR r10,[r4,#0x08]
main.s
48 adcs r10, r10, r5
49 adc r11, r11, r12 ; catch any final overflow into
50
51 ; (A_hi * B_lo)
52 umull r4, r5, r1, r2
53 adds r9, r9, r4
54 adcs r10, r10, r5
55 adc r11, r11, r12
56
57 ; (A_hi * B_hi)
58 umull r4, r5, r1, r3
59 adds r10, r10, r4
60 adc r11, r11, r5
61
62 mov r4, #0
63 mov r5, #0
64
65 ldr r4, =0x20000000
66 str r8, [r4]
67 str r9, [r4, #4]
68 str r10, [r4, #8]
69 str r11, [r4, #12]

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task4\task4.uvprojx - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Core	
R0	0xFFFFFFFF
R1	0x00000000
R2	0x00000002
R3	0x00000002
R4	0x20000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0xFFFFFFFE
R9	0xFFFFFFFF
R10	0x00000001
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x08000132
xPSR	0x01000000
N	0
Z	0
C	0
V	0
Q	0
GE	0x0
T	1
IT	Disabled
ISR	0
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	43
Sec	0.00000358
FPU	

Project | Registers

Disassembly

```
    68:                                 str r10, [r4, #8]
0x08000132 F8C4A008  STR            r10,[r4,#0x08]
    69:                                 str r11, [r4, #12]
    70:
    71:
    72:
    73:
```

main.s

```
48                  adcs r10, r10, r5
49                  adc r11, r11, r12       ; catch any final overflow into
50
51                  ; (A_hi * B_lo)
52                  umull r4, r5, r1, r2
53                  adds r9, r9, r4
54                  adcs r10, r10, r5
55                  adc r11, r11, r12
56
57                  ; (A_hi * B_hi)
58                  umull r4, r5, r1, r3
59                  adds r10, r10, r4
60                  adc r11, r11, r5
61
62                  mov r4, #0
63                  mov r5, #0
64
65                  ldr r4, =0x20000000
66                  str r8, [r4]
67                  str r9, [r4, #4]
68                  str r10, [r4, #8]
69                  str r11, [r4, #12]
70
71
72
73
74
75                  BX      LR              ; Return from main
76                  ENDP
77
78                  ALIGN
79                  END
```

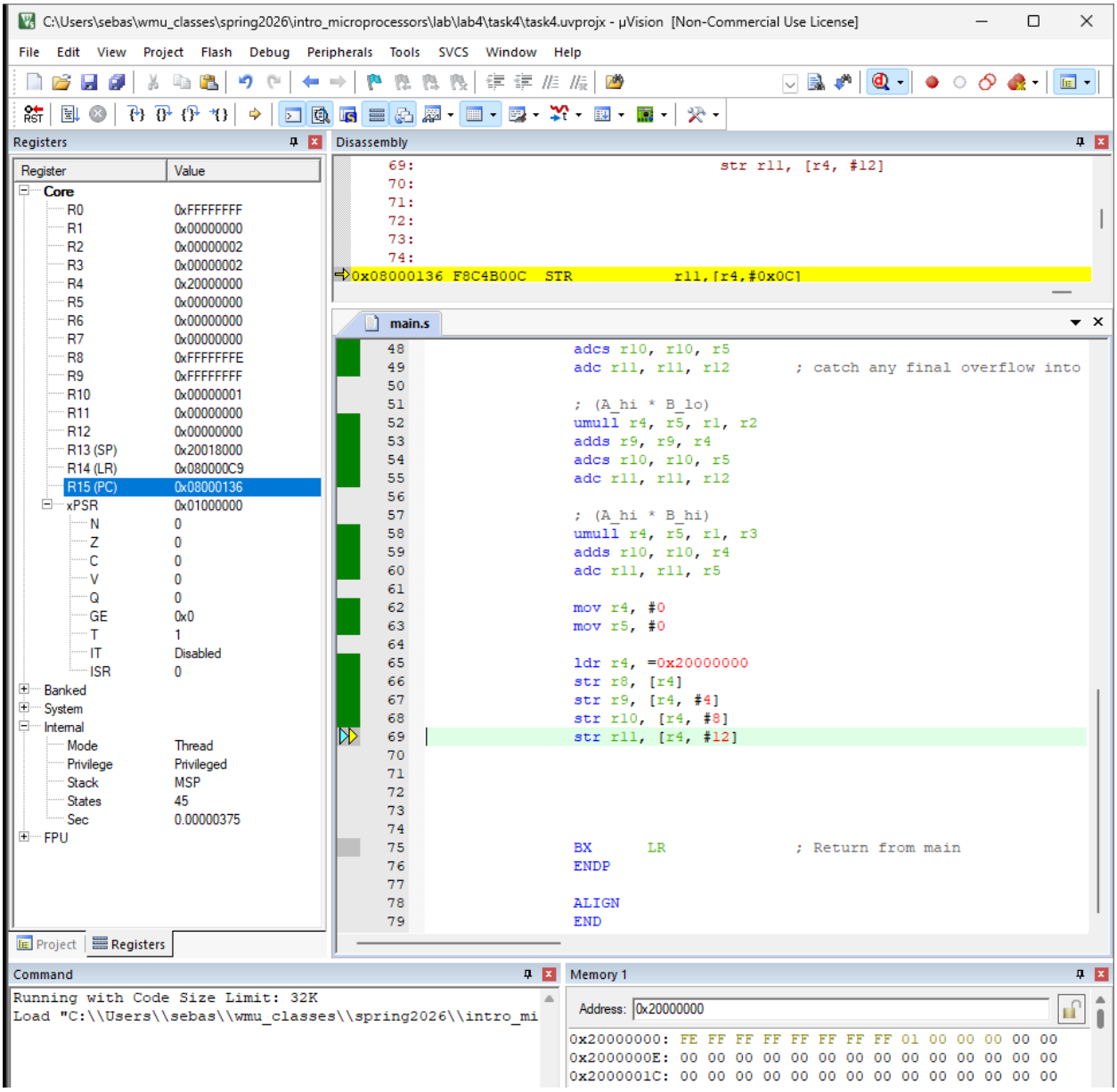
Command

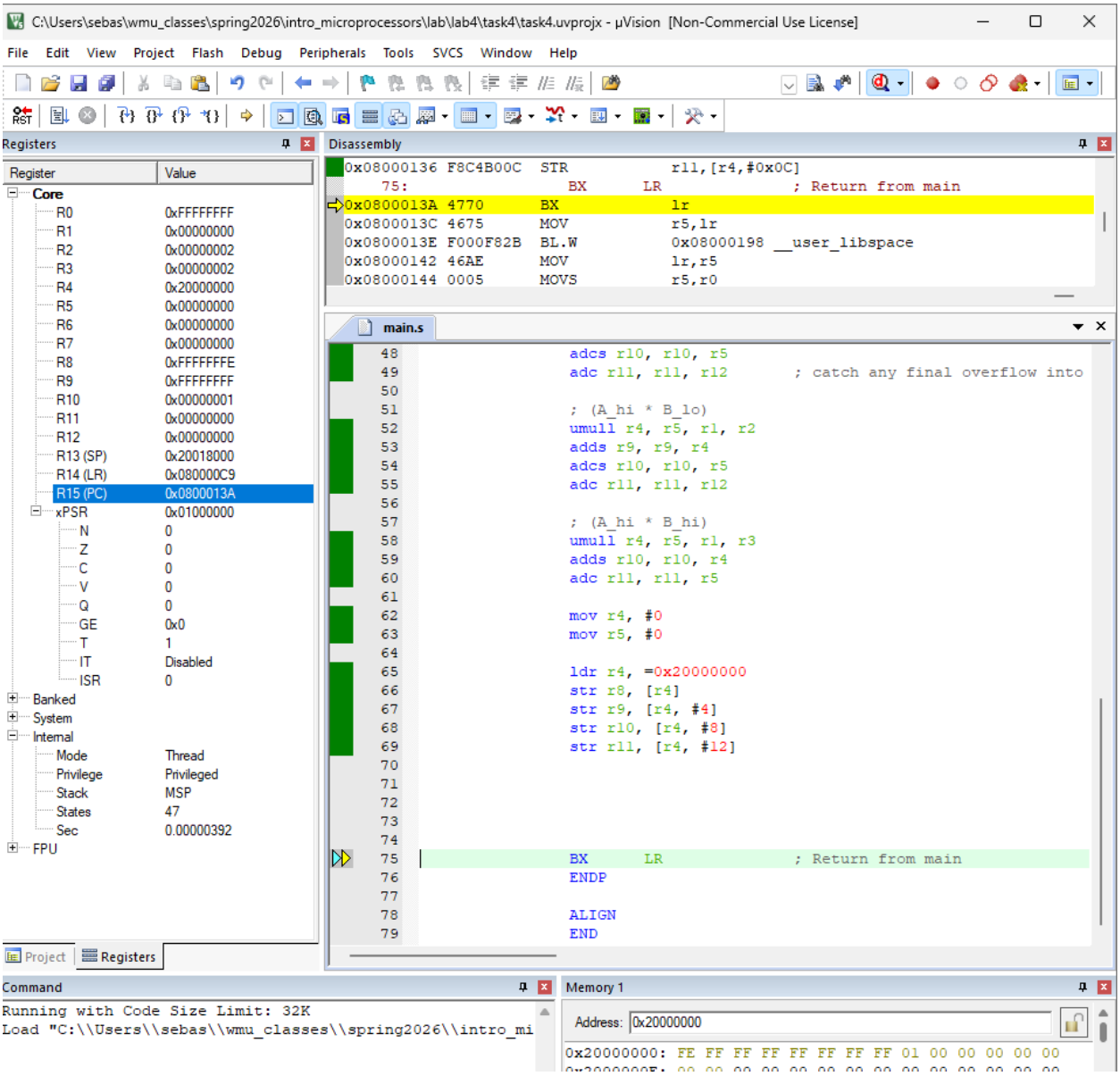
```
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
```

Memory 1

Address: 0x20000000

```
0x20000000: FE FF FF FF FF FF FF FF 00 00 00 00 00 00
0x2000000E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000001C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000002A: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task4\task4.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0xFFFFFFFF
R1 0x00000000
R2 0x00000002
R3 0x00000002
R4 0x20000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0xFFFFFFFE
R9 0xFFFFFFFF
R10 0x00000001
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x08000136
xPSR 0x01000000
N 0
Z 0
C 0
V 0
Q 0
GE 0x0
T 1
IT Disabled
ISR 0
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 45
Sec 0.00000375
FPU
Project Registers
Disassembly
69: str r11, [r4, #12]
70:
71:
72:
73:
74:
0x08000136 F8C4B00C STR r11,[r4,#0x0C]
main.s
48 adcs r10, r10, r5
49 adc r11, r11, r12 ; catch any final overflow into
50
51 ; (A_hi * B_lo)
52 umull r4, r5, r1, r2
53 adds r9, r9, r4
54 adcs r10, r10, r5
55 adc r11, r11, r12
56
57 ; (A_hi * B_hi)
58 umull r4, r5, r1, r3
59 adds r10, r10, r4
60 adc r11, r11, r5
61
62 mov r4, #0
63 mov r5, #0
64
65 ldr r4, =0x20000000
66 str r8, [r4]
67 str r9, [r4, #4]
68 str r10, [r4, #8]
69 str r11, [r4, #12]
70
71
72
73
74
75 BX LR ; Return from main
76 ENDP
77
78 ALIGN
79 END
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\sebas\\wmu_classes\\spring2026\\intro_mi
Memory 1
Address: 0x20000000
0x20000000: FE FF FF FF FF FF FF FF 01 00 00 00 00 00
0x2000000E: 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000001C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00

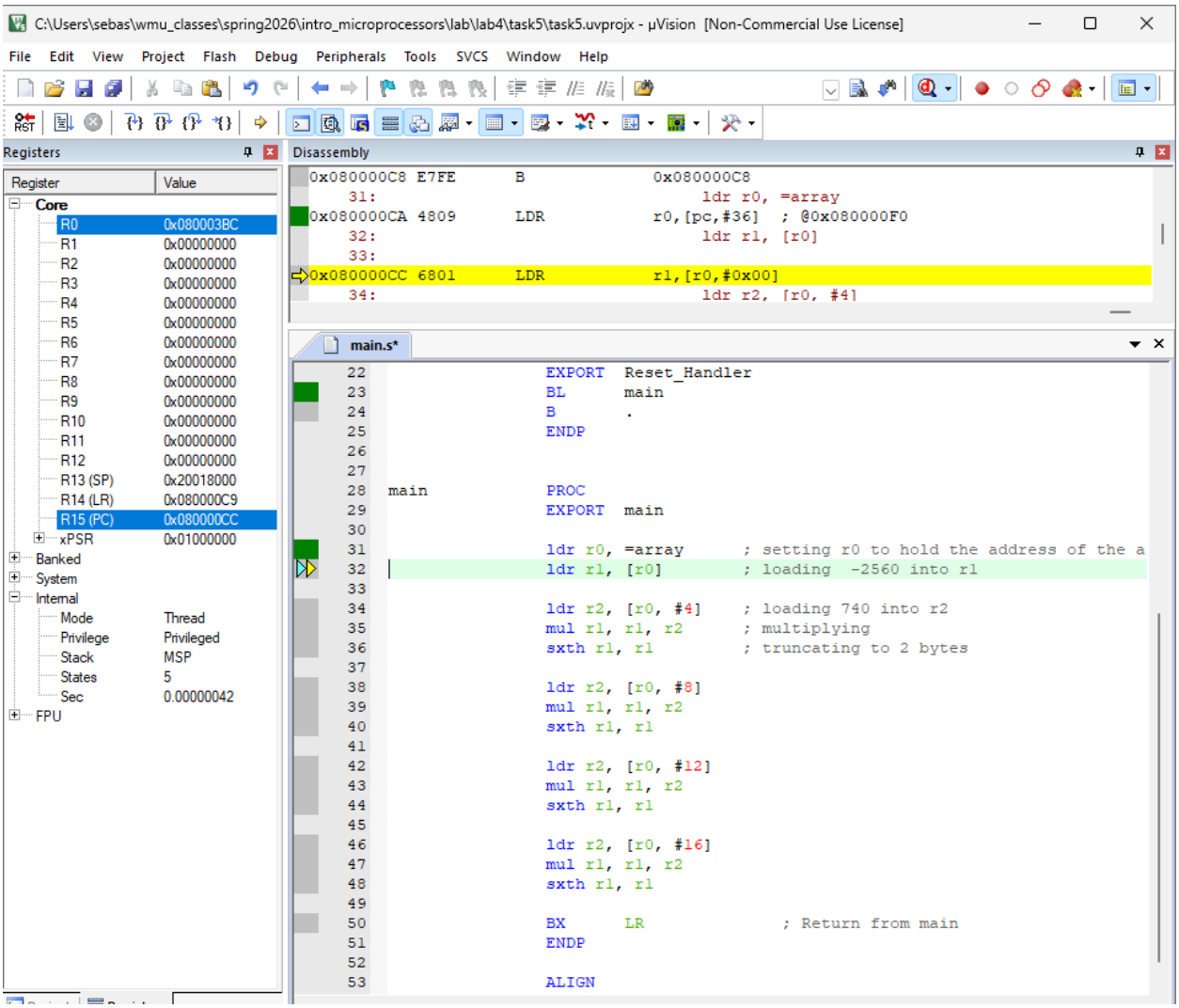


The expected result of the sum is 1FFFFFFFFFFFFFFFE

my sum is stored in little endian format taking the lsb first showing its correct

Task-5 – 1 point

1. Screenshot of the code and the result at each step.



C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task5\task5.uvprojx - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Core	
R0	0x080003BC
R1	0xFFFFF600
R2	0x00000000
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x080000CE
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	7
Sec	0.00000058
FPU	

Project | Registers

Disassembly

```
0x080000CC 6801      LDR      r1,[r0,#0x00]
    34:                           ldr r2, [r0, #4]
0x080000CE 6842      LDR      r2,[r0,#0x04]
    35:                           mul r1, r1, r2
0x080000D0 FB01F102  MUL      r1,r1,r2
    36:                           sxth r1, r1
    37:
```

main.s*

```
                BL      main
                B       .
                ENDP


main            PROC
                EXPORT  main

                ldr r0, =array      ; setting r0 to hold the address of the a
                ldr r1, [r0]        ; loading  -2560 into r1

                ldr r2, [r0, #4]    ; loading 740 into r2
                mul r1, r1, r2      ; multiplying
                sxth r1, r1         ; truncating to 2 bytes

                ldr r2, [r0, #8]
                mul r1, r1, r2
                sxth r1, r1

                ldr r2, [r0, #12]
                mul r1, r1, r2
                sxth r1, r1

                ldr r2, [r0, #16]
                mul r1, r1, r2
                sxth r1, r1

                BX      LR              ; Return from main
                ENDP

                ALIGN
                END
```

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task5\task5.uvprojx - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Core	
R0	0x080003BC
R1	0xFFFFF600
R2	0x000002E4
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x080000D0
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread

Disassembly

```
0x080000CC 6801      LDR      r1,[r0,#0x00]
    34:                           ldr r2, [r0, #4]
0x080000CE 6842      LDR      r2,[r0,#0x04]
    35:                           mul r1, r1, r2
0x080000D0 FB01F102  MUL      r1,r1,r2
    36:                           sxth r1, r1
    37:
```

main.s*

```
                BL      main
                B       .
                ENDP


main            PROC
                EXPORT  main

                ldr r0, =array      ; setting r0 to hold the address of the a
                ldr r1, [r0]        ; loading  -2560 into r1

                ldr r2, [r0, #4]    ; loading 740 into r2
                mul r1, r1, r2      ; multiplying
```

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task5\task5.uvprojx - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Core	
R0	0x080003BC
R1	0xFFE31800
R2	0x000002E4
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x080000D4
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	10
Sec	0.00000083
FPU	

Disassembly

```
0x080000D0 FB01F102  MUL     r1,r1,r2
    36:                          sxth r1, r1
    37:
0x080000D4 B209      SXTH    r1,r1
    38:                          ldr r2, [r0, #8]
0x080000D6 6882      LDR     r2,[r0,#0x08]
    39:                          mul r1, r1, r2
```

main.s*

```
23                BL      main
24                B       .
25                ENDP
26
27
28  main          PROC
29                EXPORT  main
30
31                ldr r0, =array      ; setting r0 to hold the address of the a
32                ldr r1, [r0]        ; loading  -2560 into r1
33
34                ldr r2, [r0, #4]    ; loading 740 into r2
35                mul r1, r1, r2      ; multiplying
36                sxth r1, r1         ; truncating to 2 bytes
37
38                ldr r2, [r0, #8]
39                mul r1, r1, r2
40                sxth r1, r1
41
42                ldr r2, [r0, #12]
43                mul r1, r1, r2
44                sxth r1, r1
45
```

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task5\task5.uvprojx - µVision [Non-Commercial Use License]

le Edit View Project Flash Debug Peripherals Tools SVCS Window Help

gisters

egister	Value
Core	
R0	0x080003BC
R1	0x00001800
R2	0x000002E4
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x080000D6
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	11
Sec	0.00000092

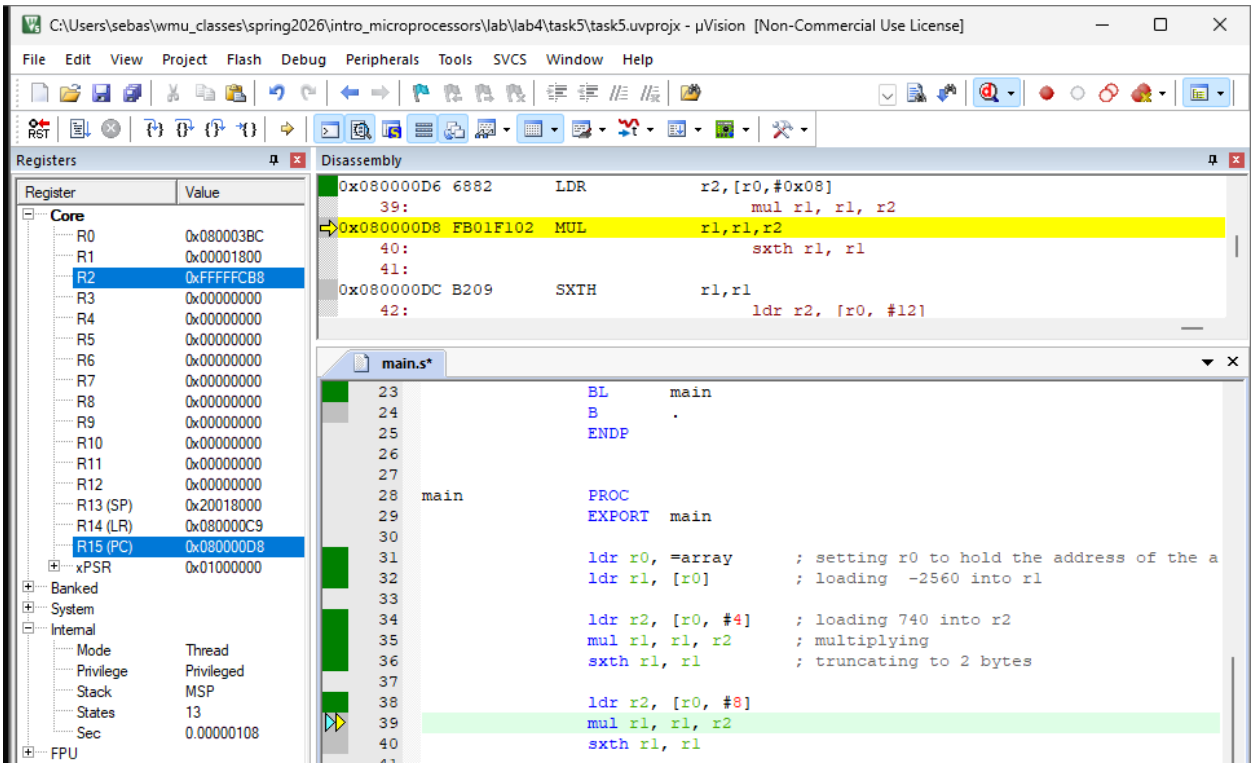
Disassembly

```
0x080000D0 FB01F102  MUL     r1,r1,r2
    36:                          sxth r1, r1
    37:
0x080000D4 B209      SXTH    r1,r1
    38:                          ldr r2, [r0, #8]
0x080000D6 6882      LDR     r2,[r0,#0x08]
    39:                          mul r1, r1, r2
```

main.s*

```
23                BL      main
24                B       .
25                ENDP
26
27
28  main          PROC
29                EXPORT  main
30
31                ldr r0, =array      ; setting r0 to hold the address of the a
32                ldr r1, [r0]        ; loading  -2560 into r1
33
34                ldr r2, [r0, #4]    ; loading 740 into r2
35                mul r1, r1, r2      ; multiplying
36                sxth r1, r1         ; truncating to 2 bytes
37
38                ldr r2, [r0, #8]
39                mul r1, r1, r2
40                sxth r1, r1
```

Multiplying with -840



C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task5\task5.uvprojx - µVision [Non-Commercial Use License]

Registers

Register	Value
Core	
R0	0x080003BC
R1	0xFFB14000
R2	0xFFFFFCB8
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x080000DC
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	14
Sec	0.00000117
FPU	

Disassembly

```
0x080000D6 6882      LDR     r2,[r0,#0x08]
    39:                          mul r1, r1, r2
0x080000D8 FB01F102  MUL     r1,r1,r2
    40:                          sxth r1, r1
    41:
0x080000DC B209      SXTH    r1,r1
    42:                          ldr r2, [r0, #12]
```

main.s*

```
23                  BL      main
24                  B       .
25                  ENDP
26
27
28  main            PROC
29                  EXPORT  main
30
31                  ldr r0, =array      ; setting r0 to hold the address of the a
32                  ldr r1, [r0]        ; loading  -2560 into r1
33
34                  ldr r2, [r0, #4]    ; loading 740 into r2
35                  mul r1, r1, r2      ; multiplying
36                  sxth r1, r1         ; truncating to 2 bytes
37
38                  ldr r2, [r0, #8]
39                  mul r1, r1, r2
40                  sxth r1, r1
41
```

C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task5\task5.uvprojx - µVision [Non-Commercial Use License]

Registers

Register	Value
Core	
R0	0x080003BC
R1	0x00004000
R2	0xFFFFFCB8
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x080000DE
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	15
Sec	0.00000125
FPU	

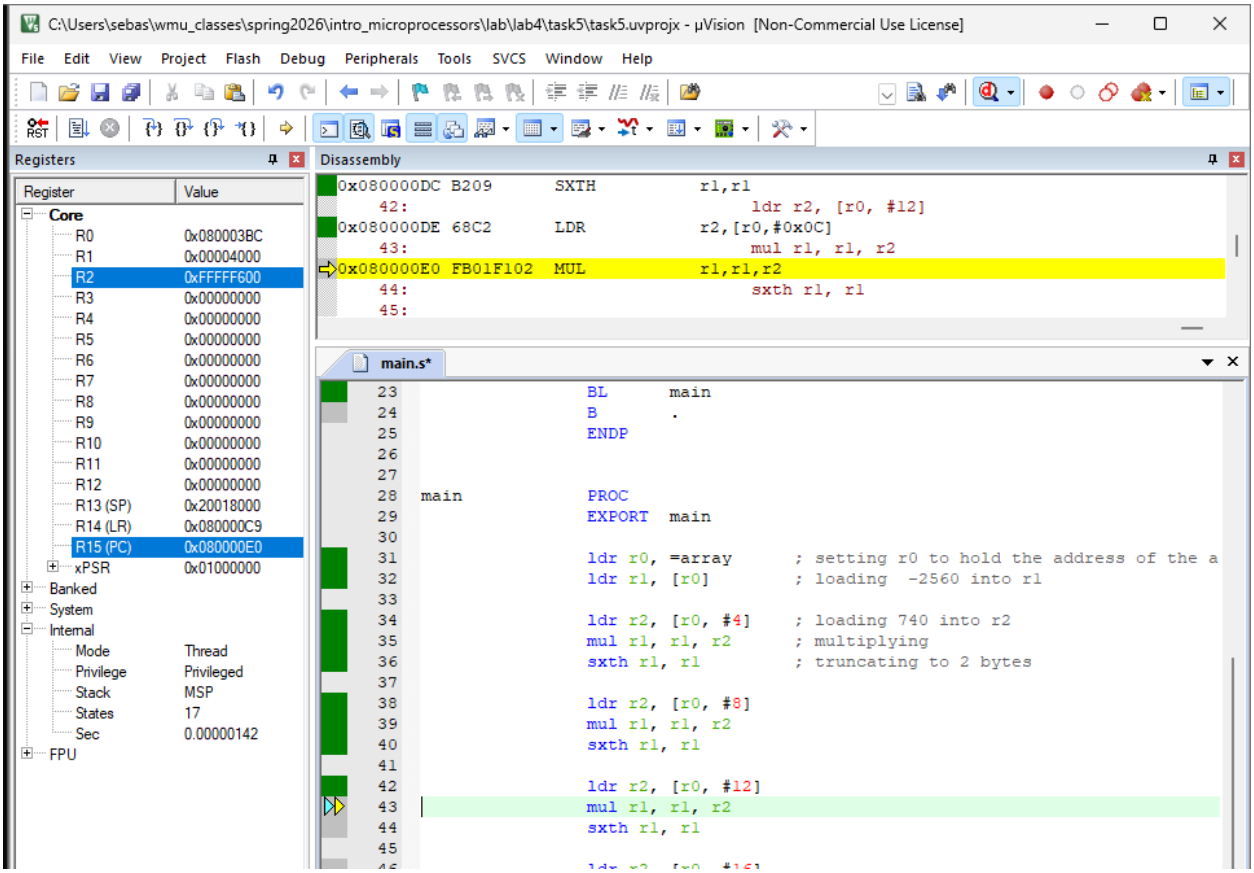
Disassembly

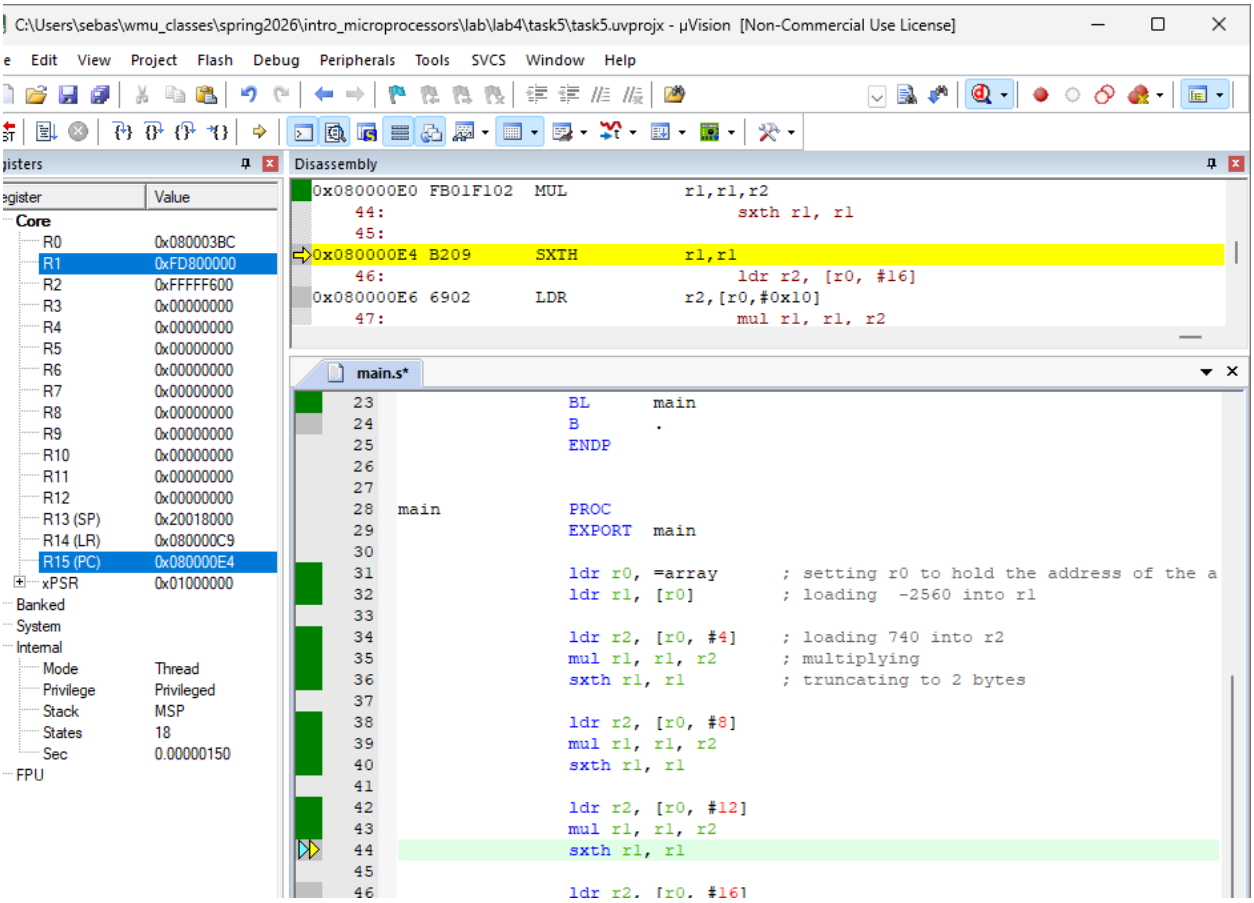
```
0x080000DC B209      SXTH    r1,r1
    42:                          ldr r2, [r0, #12]
0x080000DE 68C2      LDR     r2,[r0,#0x0C]
    43:                          mul r1, r1, r2
0x080000E0 FB01F102  MUL     r1,r1,r2
    44:                          sxth r1, r1
    45:
```

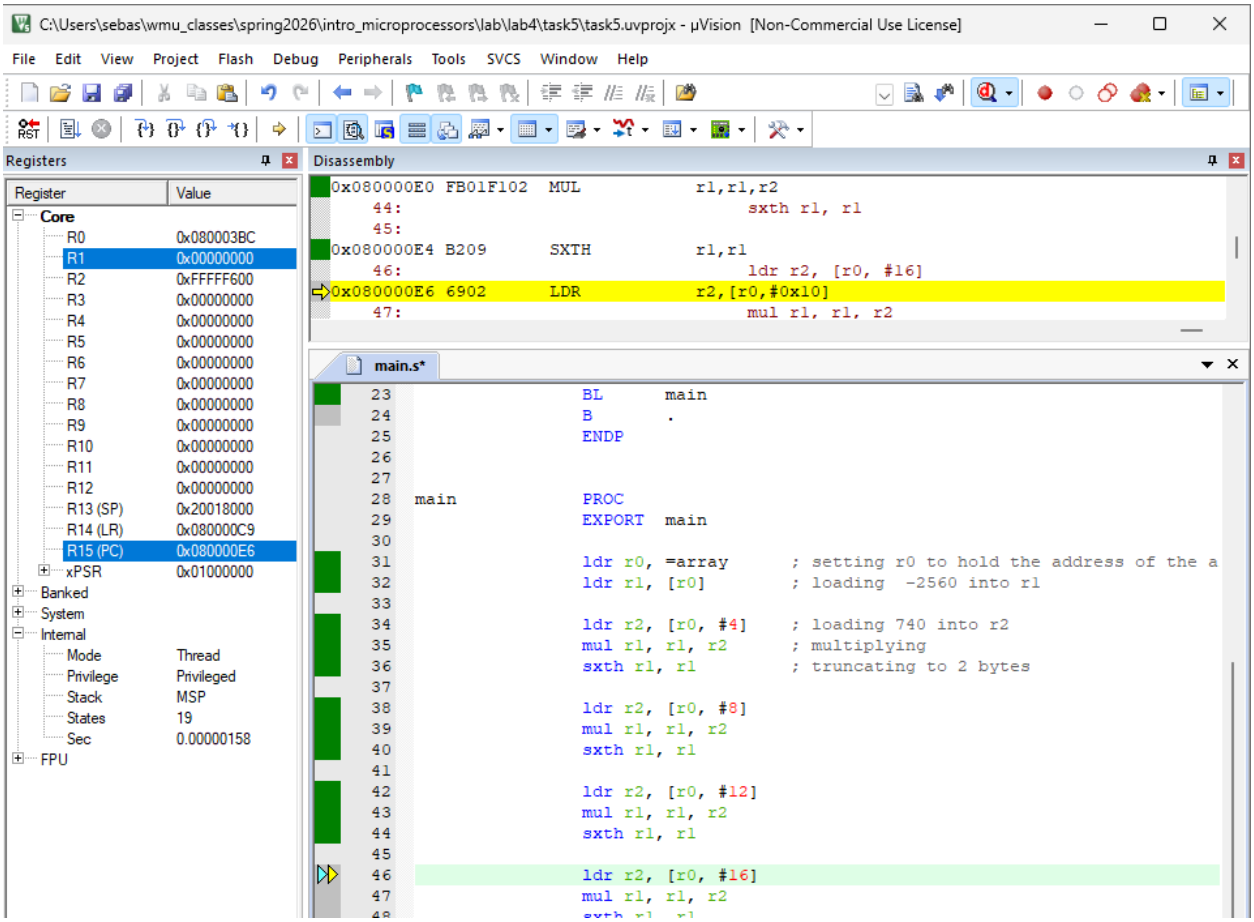
main.s*

```
23                  BL      main
24                  B       .
25                  ENDP
26
27
28  main            PROC
29                  EXPORT  main
30
31                  ldr r0, =array      ; setting r0 to hold the address of the a
32                  ldr r1, [r0]        ; loading  -2560 into r1
33
34                  ldr r2, [r0, #4]    ; loading 740 into r2
35                  mul r1, r1, r2      ; multiplying
36                  sxth r1, r1         ; truncating to 2 bytes
37
38                  ldr r2, [r0, #8]
39                  mul r1, r1, r2
40                  sxth r1, r1
41
42                  ldr r2, [r0, #12]
43                  mul r1, r1, r2
```

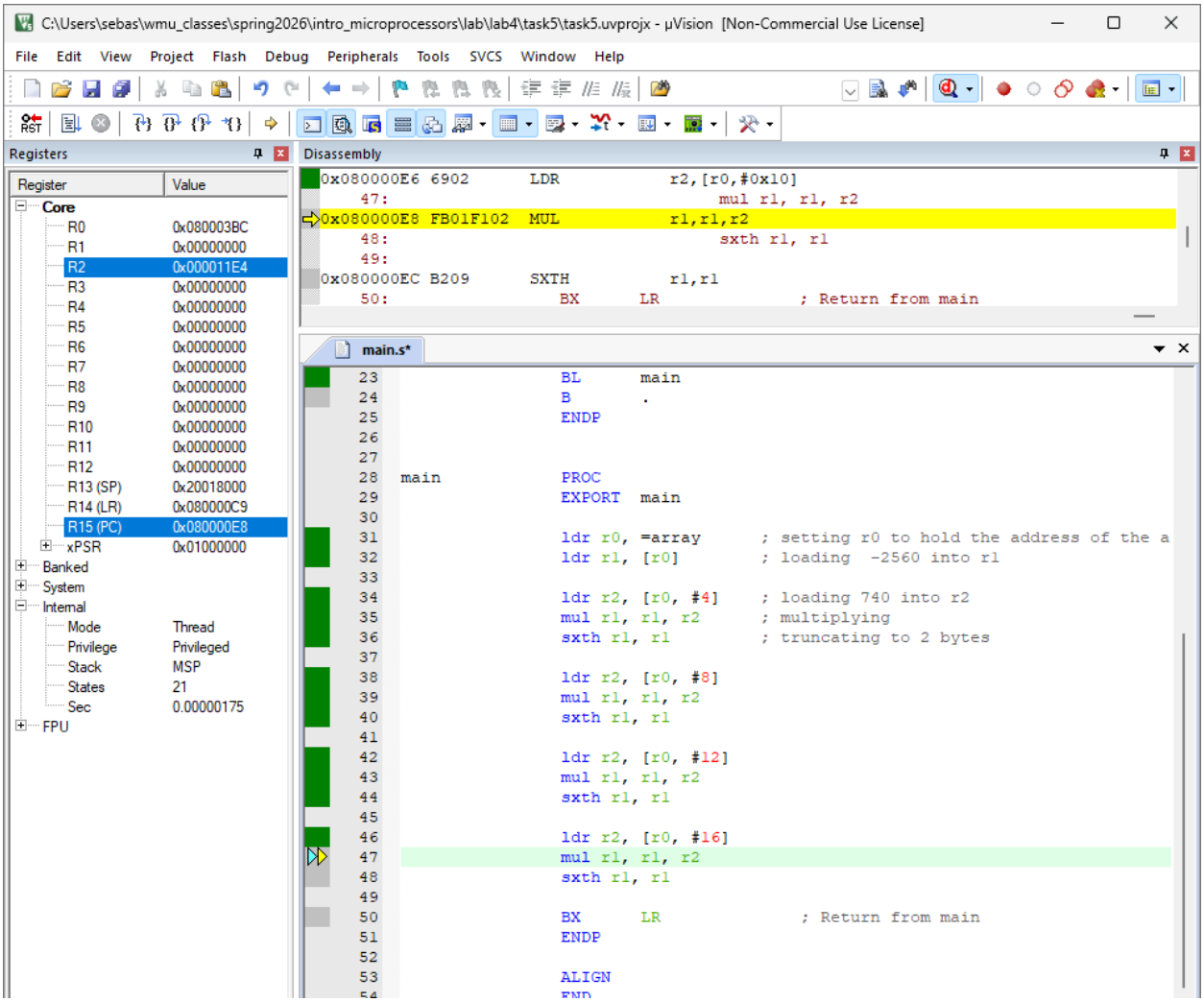
Multiplying with -2560




C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task5\task5.uvprojx - µVision [Non-Commercial Use License]
Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Disassembly
0x080000E0 FB01F102 MUL r1,r1,r2
44: sxth r1, r1
45:
0x080000E4 B209 SXTH r1,r1
46: ldr r2, [r0, #16]
0x080000E6 6902 LDR r2,[r0,#0x10]
47: mul r1, r1, r2
main.s*
BL main
B .
ENDP
main PROC
EXPORT main
ldr r0, =array ; setting r0 to hold the address of the a
ldr r1, [r0] ; loading -2560 into r1
ldr r2, [r0, #4] ; loading 740 into r2
mul r1, r1, r2 ; multiplying
sxth r1, r1 ; truncating to 2 bytes
ldr r2, [r0, #8]
mul r1, r1, r2
sxth r1, r1
ldr r2, [r0, #12]
mul r1, r1, r2
sxth r1, r1
Register Value
Core
R0 0x080003BC
R1 0xFD800000
R2 0xFFFFF600
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E4
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 18
Sec 0.00000150
FPU


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003BC
R1 0x00000000
R2 0xFFFFF600
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E6
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 19
Sec 0.00000158
FPU
Disassembly
0x080000E0 FB01F102 MUL r1,r1,r2
44: sxth r1, r1
45:
0x080000E4 B209 SXTH r1,r1
46: ldr r2, [r0, #16]
0x080000E6 6902 LDR r2,[r0,#0x10]
47: mul r1, r1, r2
main.s*
23 BL main
24 B .
25 ENDP
26
27
28 main PROC
29 EXPORT main
30
31 ldr r0, =array ; setting r0 to hold the address of the a
32 ldr r1, [r0] ; loading -2560 into r1
33
34 ldr r2, [r0, #4] ; loading 740 into r2
35 mul r1, r1, r2 ; multiplying
36 sxth r1, r1 ; truncating to 2 bytes
37
38 ldr r2, [r0, #8]
39 mul r1, r1, r2
40 sxth r1, r1
41
42 ldr r2, [r0, #12]
43 mul r1, r1, r2
44 sxth r1, r1
45
46 ldr r2, [r0, #16]
47 mul r1, r1, r2

Multiplying with 4580



File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

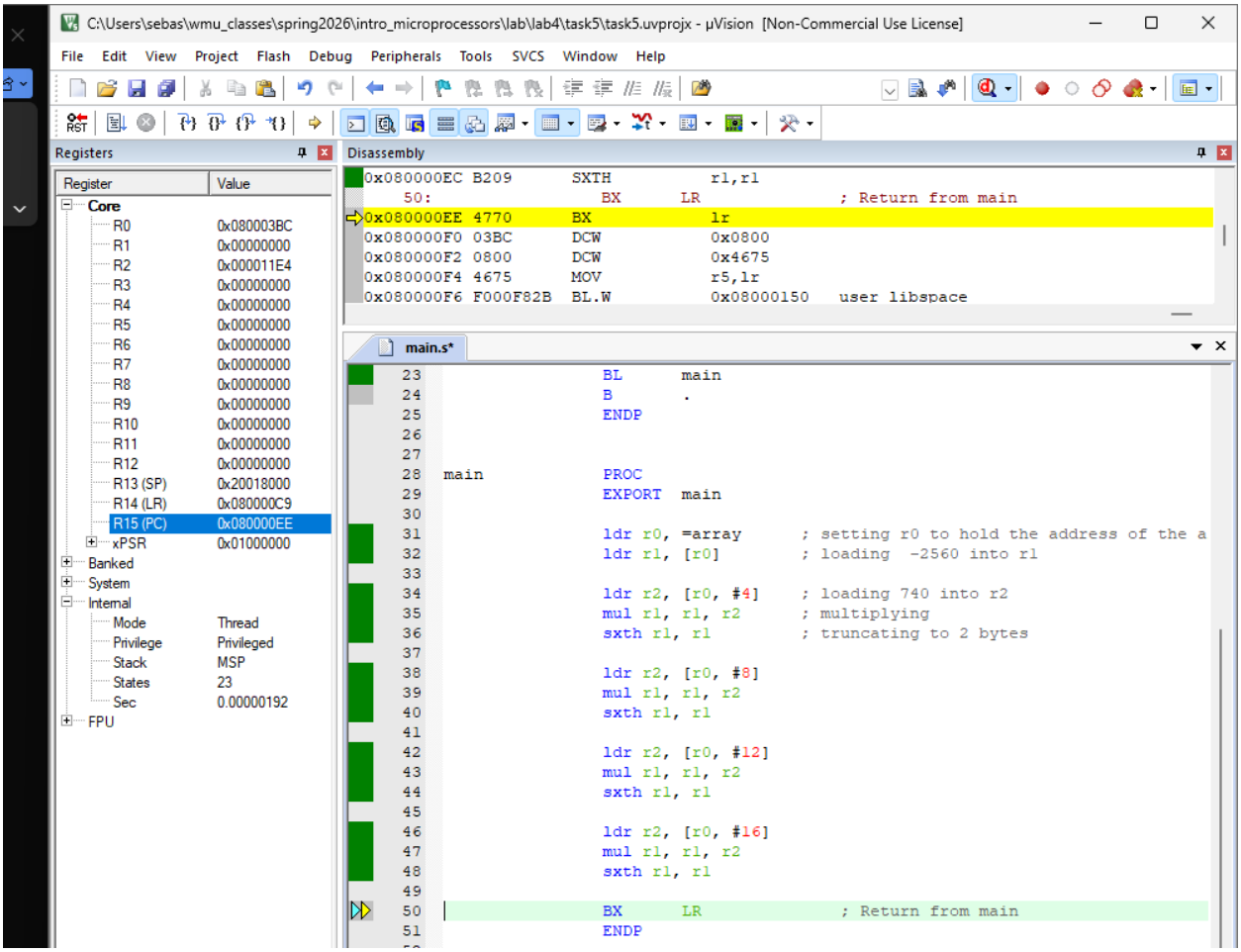
Register	Value
Core	
R0	0x080003BC
R1	0x00000000
R2	0x000011E4
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x20018000
R14 (LR)	0x080000C9
R15 (PC)	0x080000EC
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	22
Sec	0.00000183
FPU	

Disassembly

```
0x080000E6 6902      LDR        r2,[r0,#0x10]
    47:                              mul r1, r1, r2
0x080000E8 FB01F102  MUL        r1,r1,r2
    48:                              sxth r1, r1
    49:
0x080000EC B209      SXTH       r1,r1
    50:                BX      LR              ; Return from main
```

main.s*

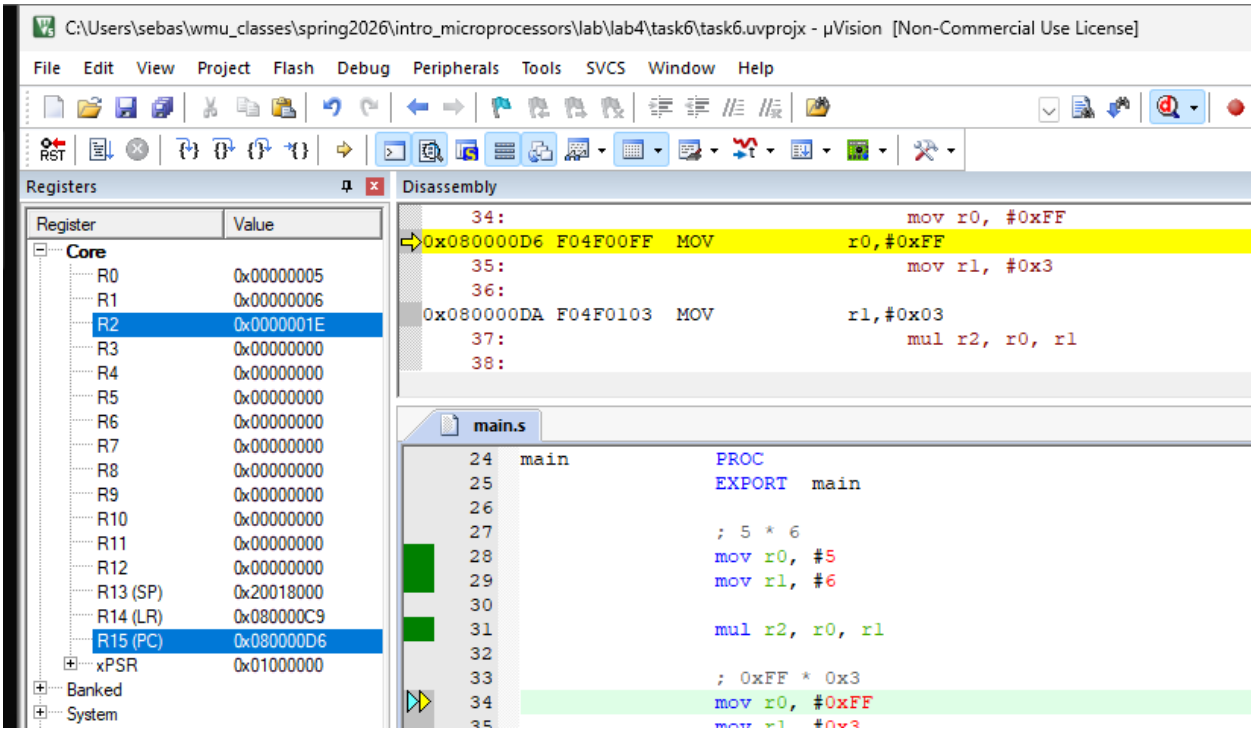
```
                BL      main
                B       .
                ENDP


main            PROC
                EXPORT  main

                ldr r0, =array      ; setting r0 to hold the address of the a
                ldr r1, [r0]        ; loading  -2560 into r1

                ldr r2, [r0, #4]    ; loading 740 into r2
                mul r1, r1, r2      ; multiplying
                sxth r1, r1         ; truncating to 2 bytes

                ldr r2, [r0, #8]
                mul r1, r1, r2
                sxth r1, r1

                ldr r2, [r0, #12]
                mul r1, r1, r2
                sxth r1, r1

                ldr r2, [r0, #16]
                mul r1, r1, r2
                sxth r1, r1

                BX      LR              ; Return from main
                ENDP
```


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task5\task5.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x080003BC
R1 0x00000000
R2 0x000011E4
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000EE
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Privilege Privileged
Stack MSP
States 23
Sec 0.00000192
FPU
Disassembly
0x080000EC B209 SXTH r1,r1
50: BX LR ; Return from main
0x080000EE 4770 BX lr
0x080000F0 03BC DCW 0x0800
0x080000F2 0800 DCW 0x4675
0x080000F4 4675 MOV r5,lr
0x080000F6 F000F82B BL.W 0x08000150 user libspace
main.s*
23 BL main
24 B .
25 ENDP
26
27
28 main PROC
29 EXPORT main
30
31 ldr r0, =array ; setting r0 to hold the address of the a
32 ldr r1, [r0] ; loading -2560 into r1
33
34 ldr r2, [r0, #4] ; loading 740 into r2
35 mul r1, r1, r2 ; multiplying
36 sxth r1, r1 ; truncating to 2 bytes
37
38 ldr r2, [r0, #8]
39 mul r1, r1, r2
40 sxth r1, r1
41
42 ldr r2, [r0, #12]
43 mul r1, r1, r2
44 sxth r1, r1
45
46 ldr r2, [r0, #16]
47 mul r1, r1, r2
48 sxth r1, r1
49
50 BX LR ; Return from main
51 ENDP

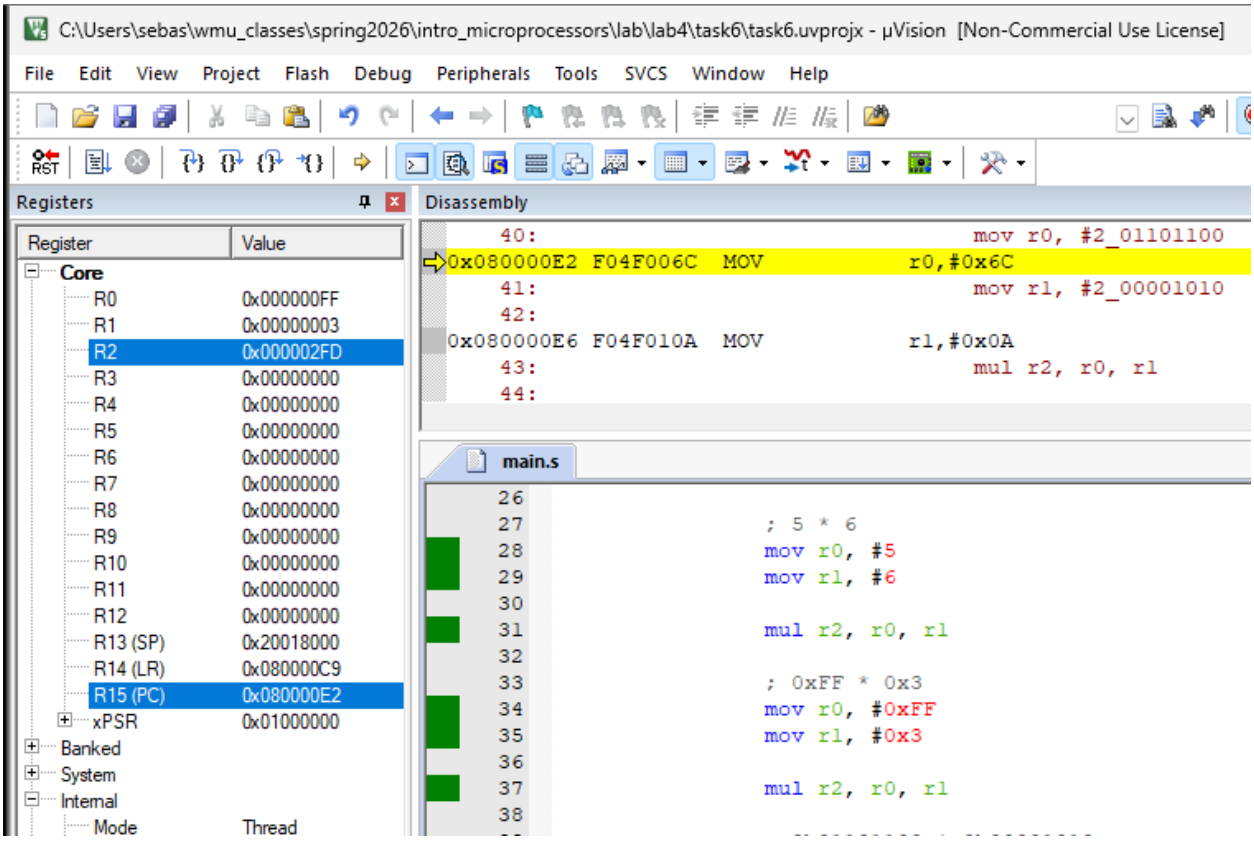
Task-6 – 1 point

1. Screenshot of the code and the result at each step.

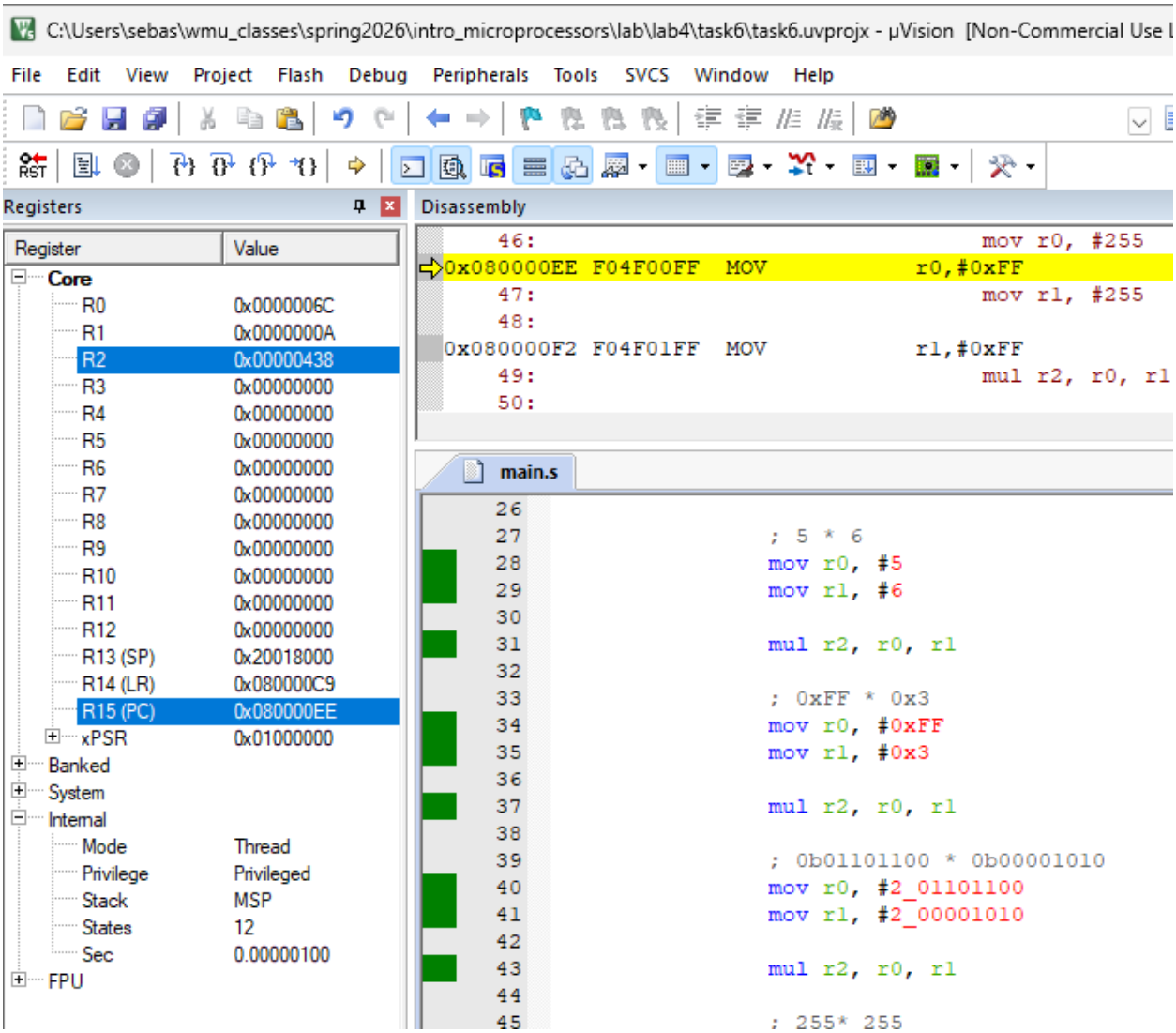
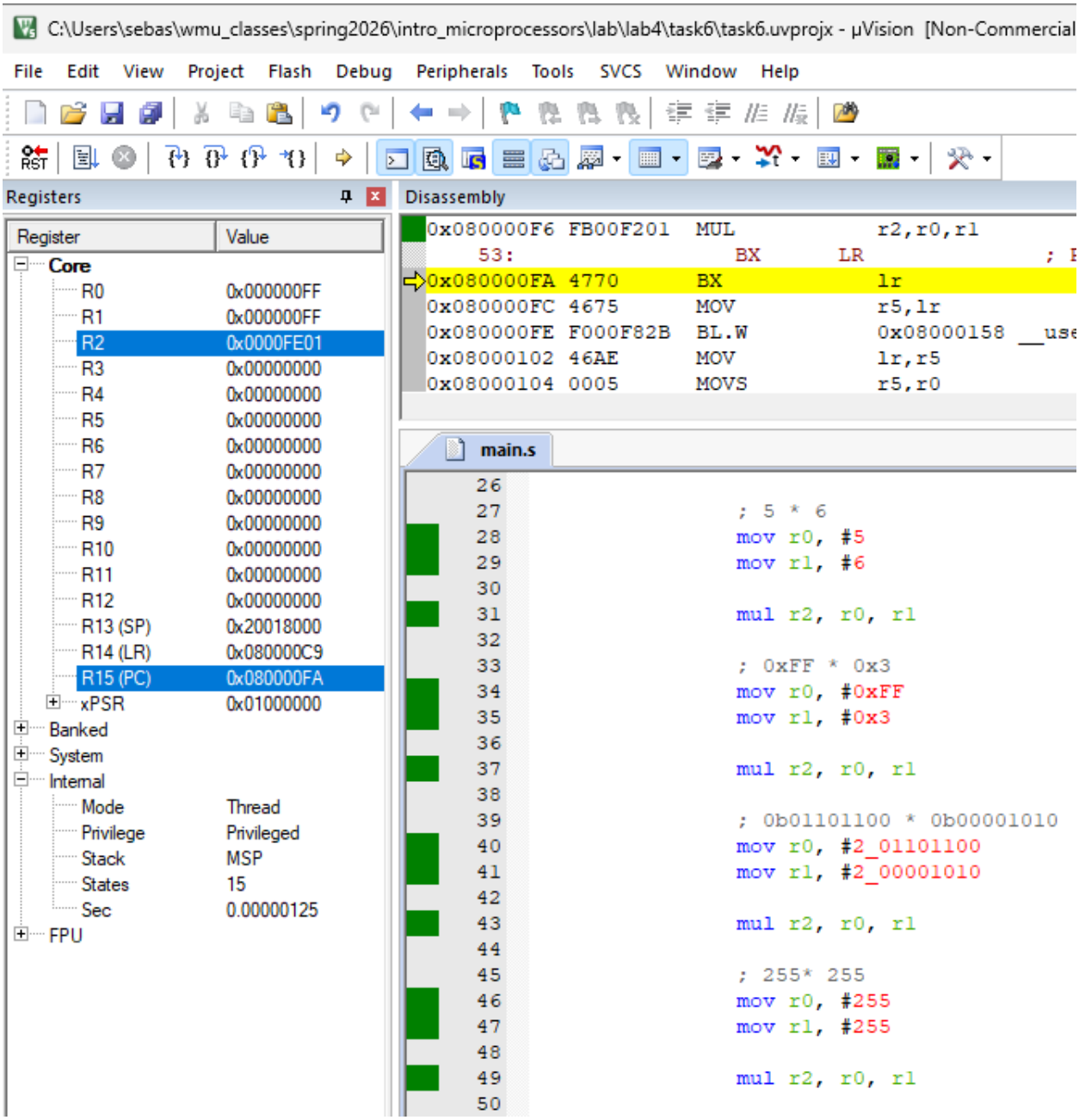
5 * 6: 36 = 1E correct



FF * 3 = 2FD


C:\Users\sebas\wmu_classes\spring2026\intro_microprocessors\lab\lab4\task6\task6.uvprojx - µVision [Non-Commercial Use License]
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Core
R0 0x000000FF
R1 0x00000003
R2 0x000002FD
R3 0x00000000
R4 0x00000000
R5 0x00000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000000
R11 0x00000000
R12 0x00000000
R13 (SP) 0x20018000
R14 (LR) 0x080000C9
R15 (PC) 0x080000E2
xPSR 0x01000000
Banked
System
Internal
Mode Thread
Disassembly
40: mov r0, #2_01101100
0x080000E2 F04F006C MOV r0,#0x6C
41: mov r1, #2_00001010
42:
0x080000E6 F04F010A MOV r1,#0x0A
43: mul r2, r0, r1
44:
main.s
26
27 ; 5 * 6
28 mov r0, #5
29 mov r1, #6
30
31 mul r2, r0, r1
32
33 ; 0xFF * 0x3
34 mov r0, #0xFF
35 mov r1, #0x3
36
37 mul r2, r0, r1
38

0b01101100 * 0b00001010 = 10000111000 = 438 in hex



255 * 255 = -511 = FE01 in hex